

Constraint-Driven Forensic Image Reconstruction Using Auditable Compressed Sensing

Benjamin Tei Partey^{a,*}, Yaw Osei Adjei^a

^a*Department of Computer Science, Kwame Nkrumah University of Science and
Technology, Kumasi, Ghana*

Abstract

Digital forensic analysis frequently recovers image evidence that is fragmented, corrupted, or partially overwritten. Deep-learning inpainting produces realistic completions, but it samples the missing pixels from a learned prior, so the output cannot be traced to the recovered bytes and carries no stated error characterisation. Both properties are obstacles to evidential use. We present Compressed Sensing Fragment Reconstruction (CSFR), a deterministic, training-free method that poses pixel-domain reconstruction as a regularised inverse problem under four constraint classes: fragment consistency, spatial continuity, intensity bounds, and frequency coherence. The solver works in a Discrete Cosine Transform (DCT) basis and carries a conditional recovery bound under the Restricted Isometry Property. We compare CSFR with five same-harness baselines (zero-fill, bilinear interpolation, Telea and Navier–Stokes partial differential equation (PDE) inpainting, and per-image dictionary learning) over four corruption families, five severity levels, and three seeds ($N = 300$ patches per cell), with a constraint ablation, paired cluster-robust inference, an iteration-budget analysis, and external validation on 176 disjoint GovDocs1 images. Under

*Corresponding author

Email address: parteybenjamintei@knust.edu.gh (Benjamin Tei Partey)

24 random pixel loss CSFR leads the strongest baseline by 1.0 to 1.4 dB; the
 25 lead survives Holm–Bonferroni correction and source-image clustering and
 26 reproduces on the external corpus. Under burst and severe half-image
 27 corruption it trails classical inpainting by up to 5.2 and 8.4 dB at the fixed
 28 600-iteration budget, and a longer solve recovers or reverses most of that
 29 deficit. The contribution is auditability. Released outputs preserve observed
 30 pixels exactly, satisfy intensity bounds, and carry a measured per-pixel
 31 provenance record; no evaluated baseline provides these as implemented.
 32 These properties support reproducibility and independent examination. Ad-
 33 missibility remains a determination for a court. All code, data, and per-seed
 34 artefacts are released.
 35 *Keywords:* Digital Forensics, Image Reconstruction, Compressed Sensing,
 36 ℓ_1 -Minimisation, Constraint-Driven Inference, Hallucination Risk,
 37 Auditability, Provenance, Forensic Validation

38 1. Introduction

39 Image evidence is one of the most frequently encountered and most evi-
 40 dentially significant artefact classes in digital forensics. Photographs, screen-
 41 shots, license-plate captures, document scans, and CCTV stills are recovered
 42 routinely from unallocated disk regions, slack space, swap files, and dam-
 43 aged storage media [1, 2, 3]. Unlike intact files retrieved through filesystem
 44 traversal, recovered image evidence is frequently *fragmented* (split across non-
 45 contiguous clusters), *corrupted* (bit-rot, burst damage, partial overwrite), or
 46 *truncated* (interrupted write cycles, partial cluster allocation). Reconstruct-
 47 ing a usable image from such evidence is therefore a precondition to almost

every downstream forensic task: visual identification, optical character recognition, metadata extraction, and human review.

1.1. The Deep-Learning Reconstruction Problem

Image reconstruction has been transformed by deep generative models. GAN-based inpainting [4], large-mask Fourier-convolutional networks [5], denoising diffusion probabilistic models [6], latent diffusion [7], and diffusion-based inpainting variants [8, 9] now produce reconstructions whose perceptual quality is, in many regimes, indistinguishable from the ground truth to human observers. These methods are powerful precisely because they replace the missing portion of an image with content drawn from a learned distribution over natural images.

In a forensic context, this strength becomes a liability. A reconstructed pixel produced by a DL inpainter is sampled from the model’s prior rather than derived from the recovered fragment bytes. The reconstructed pixel therefore constitutes a *plausible inference about what the image probably looked like* rather than a *verifiable claim about what the input fragment contains*. Recent analyses of AI-evidence admissibility have flagged this distinction as the single most consequential obstacle to deploying generative models in forensic pipelines [10, 11, 12]: the reconstructed content cannot be traced to a verifiable algebraic transformation of the input, and so cannot withstand a Daubert challenge to its reliability [13] or a defence argument that the investigation was conducted with methodologically unsound tools [14].

70 *1.2. The Reconstruction Gap in Forensic Tooling*

71 The reconstruction options available to forensic practitioners span a spec-
72 trum whose endpoints are unsatisfactory in opposite ways:

- 73 1. **Classical signature-based carvers** (PhotoRec, Foremost, Scalpel)
74 recover whole files by header-footer matching but perform *no* recon-
75 struction on corrupted or fragmented bytes; corrupted regions remain
76 corrupted on output [15, 3].
- 77 2. **Classical interpolation and inpainting** (bilinear, PDE diffusion,
78 per-image dictionary learning) do reconstruct corrupted regions and
79 draw only on the image’s own content, which places them closer to
80 forensic acceptability than learned priors; they are the strongest base-
81 lines in this paper. But their standard implementations emit no prove-
82 nance record, carry no closed-form recovery characterisation, and give
83 no hard guarantee that observed pixels are returned unaltered.
- 84 3. **Deep-learning inpainters** (generative adversarial network (GAN)-
85 based, diffusion-based) reconstruct with high perceptual fidelity but
86 produce content sampled from a learned prior, which is, in the formal
87 sense, hallucinated [4, 5, 6, 7, 8].

88 The classical middle already supplies non-deep-learning reconstruction.
89 What no available method supplies is reconstruction that is simultaneously
90 bounded (a stated recovery characterisation), evidence-preserving (observed
91 pixels returned unaltered by construction), and traceable (a released per-
92 pixel provenance record). The present paper targets that combination and
93 positions CSFR against the classical middle, which is the strongest of the
94 three regimes for this purpose.

95 1.3. Motivating Evidence from Prior Work

96 Bounded reconstruction is an operational necessity. Fragment classifica-
97 tion and carving pipelines degrade sharply when their inputs are corrupted:
98 file-carving studies document that fragmentation and partial overwrites are
99 pervasive in field-recovered media [3, 15], and analyses of AI-assisted forensic
100 tooling flag both silent performance collapse and miscalibrated confidence on
101 damaged inputs as central reliability risks [10, 11, 12]. A reconstruction stage
102 that repairs what can be repaired, and explicitly refuses what cannot, is there-
103 fore the structural precondition for any trustworthy downstream analysis of
104 corrupted image fragments.

105 1.4. Research Question

106 The paper addresses one question:

107 *Can a deterministic, constraint-driven sparse-recovery procedure*
108 *reconstruct corrupted image fragments at a quality competitive*
109 *with classical inpainting, while emitting the error characterisa-*
110 *tion and per-pixel provenance record that evidential use requires*
111 *and conventional inpainting pipelines do not provide?*

112 1.5. Contributions

113 **1. Constraint-driven reconstruction framework.** Forensic image re-
114 construction is formalised as a regularised inverse problem under four
115 constraint families with forensic interpretations (spatial continuity, in-
116 tensity bounds, frequency coherence, fragment consistency), with the
117 conditions under which recovery is bounded and a proof that fragment
118 consistency is hard-enforced by construction (Section 3).

- 119 **2. CSFR solver.** A six-stage deterministic, training-free solver (ingest,
120 alignment, constraint formulation, ℓ_1 -minimisation in a sparsifying ba-
121 sis, assembly, traceability audit) that emits a per-pixel provenance
122 record linking every output pixel to the observed inputs that deter-
123 mine it, computed by implicit differentiation at the solver’s stationarity
124 condition (Section 4).
- 125 **3. Forensic evaluation protocol.** Five metrics beyond perceptual fi-
126 delity: PSNR, SSIM, Constraint Violation Rate, Hallucination Risk
127 Proxy, and Traceability Score, with definitions and reporting conven-
128 tions specified (Section 5).
- 129 **4. Same-harness evaluation with inference.** A released comparison
130 against five classical baselines with a constraint ablation, paired cluster-
131 robust tests with effect sizes, an iteration-budget analysis, and external
132 validation on a disjoint corpus. The regimes where CSFR loses are
133 reported as fully as those where it wins (Sections 6, 6.10).

134 The remainder of the paper is organised as follows. Section 2 surveys
135 related work in classical reconstruction, forensic image analysis, and the DL
136 methods cited for contrast. Section 3 formalises the reconstruction problem
137 and defines the four constraint families. Section 4 describes the CSFR solver.
138 Section 5 specifies the experimental protocol and metrics. Section 6 reports
139 results from a real 2-D image-patch CF $\times$ L sweep. Section 7 interprets the
140 findings for forensic practitioners. Section 8 states limitations. Section 9
141 concludes.

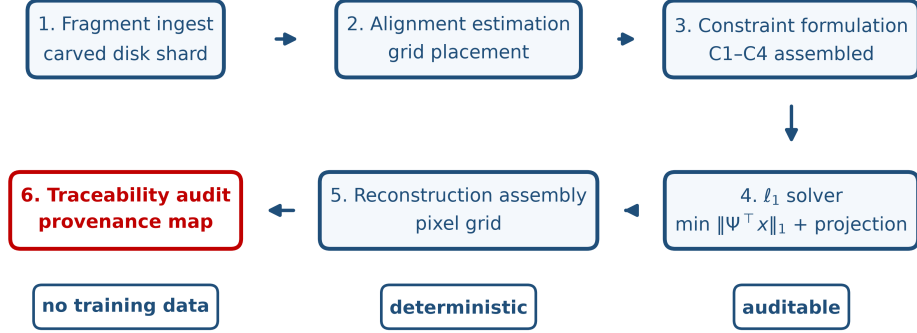


Figure 1: The CSFR constraint-driven reconstruction pipeline. A recovered fragment passes through six deterministic stages: ingest, alignment estimation, constraint formulation, ℓ_1 -minimisation in a sparsifying basis, reconstruction assembly, and traceability audit. The traceability audit emits a per-pixel provenance map that is the artefact required for chain-of-custody documentation under the Daubert standard.

142 2. Related Work

143 2.1. Compressed Sensing and Sparse Recovery

144 Compressed sensing (CS) establishes that signals which are sparse in some
 145 basis can be recovered exactly from a small number of incoherent linear mea-
 146 surements via convex ℓ_1 -minimisation [16, 17, 18]. The Restricted Isometry
 147 Property (RIP) [19] provides a sufficient condition under which the recovery
 148 is unique and stable to noise. CS is therefore a natural mathematical scaffold
 149 for forensic reconstruction: image fragments are sparse in DCT or wavelet
 150 bases [20], the corruption operator is linear, and the recovery has a closed
 151 analytical form that is auditable.

152 The application of CS to image reconstruction in non-forensic contexts is
 153 mature: total-variation regularisation [21], Tikhonov regularisation [22], and

154 dictionary-based sparse coding [23, 24, 25] are standard tools. What distin-
155 guishes the present work is the deliberate restriction of the reconstruction
156 objective to constraints whose violation has a forensic interpretation, and
157 the explicit rejection of any data-driven prior whose origins lie outside the
158 input fragment.

159 2.2. Classical Image Inpainting

160 Bertalmio et al. [26] introduced PDE-based diffusion inpainting; Criminisi
161 et al. [27] extended it with exemplar-based texture synthesis. Both methods
162 produce visually pleasing fills but borrow content from surrounding image re-
163 gions; in forensic terms, this is a controlled form of hallucination because the
164 filled content originates from the same image rather than from an unverified
165 external prior. Classical inpainting is therefore closer to forensic admissibility
166 than DL inpainting, but it is not equivalent: the filled pixels remain derived
167 from a heuristic rather than from a mathematically bounded transformation
168 of the input fragment.

169 2.3. Deep Learning Inpainting and Diffusion (Contrast Only)

170 We cite GAN inpainting [4], Fourier-convolutional large-mask inpaint-
171 ing [5], denoising diffusion [6], latent diffusion [7], RePaint [8], pluralistic com-
172 pletion [9], and recent latent- and gradient-guided diffusion inpainters [28, 29]
173 only to make the contrast explicit: each of these methods produces output
174 pixels that are sampled from a learned prior. The claim we make about them
175 is deliberately narrow. We do not claim that deep inpainting is unsuitable
176 for forensic work in general; for triage, prioritisation, and lead generation it
177 may be the better tool, and this paper does not argue otherwise. The claim

178 is confined to the evidential use case, where a reconstruction or a conclusion
 179 drawn from it is tendered to a finder of fact, sampling from a learned prior
 180 forfeits two properties that evidential use asks for: a closed error characterisa-
 181 tion, and a derivation of each output pixel from the input under examination.
 182 Section 7.1 develops this distinction. These methods are cited rather than
 183 run because reproducing them under this harness would require the learned
 184 priors whose evidential status is precisely what is at issue, and a perceptual-
 185 quality comparison against them would not settle the question the paper asks;
 186 we note in Section 8 that this makes the comparison structurally asymmetric.

187 2.4. Forensic File Carving

188 File carving research [3, 15, 30, 31] (including statistical content-based
 189 fragment identification [32] built on entropy measures [33]) has historically
 190 focused on *reassembly* (locating and ordering fragments) rather than *recon-*
 191 *struction* (filling corrupted bytes). The present work occupies the latter niche
 192 and is complementary to existing carving systems: CSFR is intended to op-
 193 erate on individual fragments after a carver has located them.

194 2.5. Forensic Admissibility Frameworks

195 The Daubert standard [13] and SWGDE best-practice guidance [34]
 196 jointly establish that algorithmic forensic tools must have known, quan-
 197 tifiable error rates and must produce outputs that are auditable. Farid’s
 198 survey of image forgery detection [35] provides the canonical reference for
 199 the legal context. Work on reconstruction from incomplete measurements
 200 has established that hallucination is not an incidental defect of particular
 201 architectures but a structural property of regularised inversion: Bhadra et

202 al. [10] show that a reconstruction can be simultaneously data-consistent
 203 and contain features absent from the object, locating the difficulty in the
 204 component of the estimate lying in the null space of the forward operator,
 205 and Tivnan et al. [36] develop a corresponding image-quality index. In
 206 parallel, the digital-forensics literature has begun to formalise tool error and
 207 its disclosure [11, 12]. The present work sits at that intersection. It does
 208 not eliminate the null-space problem, since no reconstruction method can:
 209 the missing pixels are by definition unconstrained by the data. It constrains
 210 that component explicitly through C2–C4 and records what determined it,
 211 so that the inferred part of the output is bounded and inspectable.

212 **3. Problem Formulation**

213 *3.1. Notation*

214 Let $X \in \mathbb{R}^{H \times W}$ denote a ground-truth image with H rows and W columns,
 215 vectorised as $x \in \mathbb{R}^N$ with $N = HW$. Let $\Omega \subseteq \{1, \dots, N\}$ denote the set of
 216 observed (intact) pixel indices, and $\Omega^c = \{1, \dots, N\} \setminus \Omega$ the set of corrupted
 217 or missing pixel indices. The observed (corrupted) image $y \in \mathbb{R}^N$ satisfies
 218 $y_i = x_i$ for $i \in \Omega$ and is undefined (or arbitrarily set to zero) for $i \in \Omega^c$.
 219 Let $\Psi \in \mathbb{R}^{N \times N}$ denote a sparsifying basis (e.g., the two-dimensional Discrete
 220 Cosine Transform (DCT), or a wavelet basis) such that the coefficient vector
 221 $\alpha = \Psi^\top x$ is approximately κ -sparse with $\kappa \ll N$. We use the vectorised form
 222 $x \in \mathbb{R}^N$ for the analysis and the equivalent patch form $\text{mat}(x) \in \mathbb{R}^{H \times W}$ for
 223 the solver; the two are related by $x = \text{vec}(\text{mat}(x))$. For the separable 2-D
 224 DCT, $\Psi^\top x = \text{vec}(D \text{mat}(x) D^\top)$ where $D \in \mathbb{R}^{H \times H}$ (with $H = W$ for square
 225 patches) is the 1-D orthonormal DCT matrix, i.e. $\Psi = D \otimes D$; the algorithm

226 of Section 4 works in the patch form with D , and the analysis here works in
 227 the vectorised form with Ψ .

228 Let $M \in \{0, 1\}^{N \times N}$ denote the diagonal sampling mask with $M_{ii} = 1$ for
 229 $i \in \Omega$ and $M_{ii} = 0$ otherwise. The forward (corruption) model is:

$$y = Mx + \eta \quad (1)$$

230 where $\eta \in \mathbb{R}^N$ is bounded noise with $\|\eta\|_2 \leq \varepsilon$.

231 **Domain of reconstruction.** CSFR operates in the *decoded pixel do-*
 232 *main*: the input is the pixel array obtained after any applicable container
 233 decoding of the carved fragment, and the output is a reconstructed pixel ar-
 234 ray. CSFR does not claim byte-perfect recovery of the original encoded file
 235 (e.g., the exact JPEG bitstream); recovering an encoded bitstream whose
 236 decode matches the reconstruction is a distinct problem outside this paper’s
 237 scope, and all traceability claims refer to the pixel-domain transformation
 238 chain.

239 3.2. Reconstruction Objective

240 The constraint-driven reconstruction objective is:

$$\hat{x} = \arg \min_{x \in \mathbb{R}^N} \|\Psi^\top x\|_1 \quad \text{s.t.} \quad \|Mx - y\|_2 \leq \varepsilon, \quad x \in \mathcal{F}_{\text{constraints}} \quad (2)$$

241 where $\mathcal{F}_{\text{constraints}}$ is the feasibility set induced by the four constraint families
 242 defined below. The ℓ_1 term enforces sparsity in Ψ ; the data-fidelity constraint
 243 enforces consistency with the observed fragment bytes; the feasibility set
 244 enforces the forensic constraints.

245 3.3. Constraint Family C1: Fragment Consistency (Hard)

246 **Definition.** For all observed indices $i \in \Omega$, $\hat{x}_i = y_i$.

247 **Forensic interpretation.** A reconstruction must not alter pixels that
 248 were directly observed in the recovered fragment. This is the central forensic
 249 constraint: observed bytes are evidence and are immutable. CSFR enforces
 250 this constraint as a hard equality (not a penalised soft constraint) so that
 251 the data-fidelity tolerance ε applies only to noise on the observation channel,
 252 never to substantive alteration of intact pixels.

253 3.4. Constraint Family C2: Spatial Continuity

254 **Definition.** For all interior pixels i , the local total variation is bounded:

$$\text{TV}_{\text{local}}(\hat{x}, i) = \sum_{j \in \mathcal{N}(i)} |\hat{x}_i - \hat{x}_j| \leq \tau_{\text{TV}}(i) \quad (3)$$

255 where $\mathcal{N}(i)$ is the 4-neighbourhood of i and $\tau_{\text{TV}}(i)$ is a position-dependent
 256 bound estimated from the local statistics of $\Omega \cap \mathcal{N}(i)$.

257 **Forensic interpretation.** Real-world images exhibit piecewise smooth-
 258 ness; a reconstruction that introduces high-frequency discontinuities at the
 259 boundary of Ω and Ω^c is implausible and indicative of unbounded inference.

260 3.5. Constraint Family C3: Intensity Bounds

261 **Definition.** For all $i \in \Omega^c$,

$$\hat{x}_i \in [\ell_i, u_i] \quad (4)$$

262 where ℓ_i, u_i are derived from the empirical minimum and maximum of $y_{\Omega \cap \mathcal{N}(i)}$
 263 plus a bounded slack.

264 **Forensic interpretation.** A reconstructed pixel cannot exceed the dy-
 265 namic range of its observed neighbourhood; an inferred pixel value of 0 or
 266 255 in a region whose neighbours are all near grey-level 128 would be a hal-
 267 lucination.

268 3.6. Constraint Family C4: Frequency Coherence

269 **Definition.** The high-frequency tail of $\Psi^\top \hat{x}$ is bounded:

$$\sum_{h \in \mathcal{H}} (\Psi^\top \hat{x})_h^2 \leq \rho \|\Psi^\top \hat{x}\|_2^2 \quad (5)$$

270 where $\mathcal{H}$ indexes the high-frequency basis coefficients and $\rho \in (0, 1)$ is a
 271 coherence parameter.

272 **Forensic interpretation.** Natural images concentrate energy in low-
 273 frequency basis components; a reconstruction with anomalously high high-
 274 frequency energy indicates either noise amplification or imagined structure.

275 3.7. Recovery Conditions

276 Under the standard CS Restricted Isometry Property assumption on
 277 $M\Psi$ [19], the unique minimiser $\hat{x}$ of Eq. (2) satisfies, for any κ -sparse ground
 278 truth $\alpha^\star = \Psi^\top x^\star$:

$$\|\hat{x} - x^\star\|_2 \leq K_0 \varepsilon + K_1 \frac{\|\alpha^\star - \alpha_\kappa^\star\|_1}{\sqrt{\kappa}} \quad (6)$$

279 where $\alpha_\kappa^\star$ is the best κ -term approximation and K_0, K_1 are constants depend-
 280 ing on the RIP constant of $M\Psi$. The recovery is therefore *bounded* (the
 281 reconstruction error is provably finite) and *stable* (small perturbations of y
 282 produce small perturbations of $\hat{x}$). These two properties together are the
 283 formal counterpart of the forensic requirement that the reconstruction be

284 reproducible and auditable. Two qualifications bound this guarantee. First,
 285 the bound is *conditional*: it holds only when the sensing operator $M\Psi$ actu-
 286 ally satisfies the RIP at the relevant sparsity level κ , a condition that severe
 287 or adversarially structured corruption can violate (Section 8); verifying RIP
 288 for a given mask is itself computationally hard, so the bound should be
 289 read as a sufficient-condition guarantee, not a certificate attached to each
 290 reconstruction. Second, the bound constrains *numerical* error, not *semantic*
 291 correctness: a reconstruction within the error bound may still be interpreted
 292 incorrectly by a downstream examiner, and nothing in Eq. (2) certifies the
 293 evidential meaning of the recovered content.

294 4. The CSFR Reconstruction Framework

295 This section describes the six-stage CSFR solver pipeline (Figure 1).

296 4.1. Stage 1: Fragment Ingest

297 The carver provides one or more byte sequences $\{f^{(1)}, \dots, f^{(F)}\}$ identified
 298 as belonging to a single recoverable image, together with a corruption mask
 299 $M^{(f)}$ per fragment indicating which bytes are intact and which are corrupted.
 300 CSFR makes no assumption about how the carver located these fragments;
 301 it operates on whatever the carver supplies.

302 4.2. Stage 2: Alignment Estimation

303 Each fragment is mapped to its position in the canonical pixel grid via
 304 header-derived offsets (for JPEG, the index of the minimum coded unit
 305 (MCU); for raw bitmaps, the byte offset divided by the row stride). Where
 306 headers are absent or corrupted, alignment is estimated by cross-correlation

307 of the fragment edge statistics with a low-resolution reconstruction proposal
 308 (the zero-fill composite of all fragments already placed, downsampled $4\times$,
 309 which requires no learned prior). Alignment uncertainty is propagated for-
 310 ward as additional erasure indices in Ω^c .

311 4.3. Stage 3: Constraint Formulation

312 Given the aligned fragments and the corruption mask, CSFR instantiates
 313 the four constraint families C1–C4 of Section 3. The bounds $\tau_{\text{TV}}(i)$, $[\ell_i, u_i]$,
 314 and ρ are computed from the observed neighbourhood statistics of each $i \in$
 315 Ω^c , ensuring that the feasibility set $\mathcal{F}_{\text{constraints}}$ depends only on the input
 316 fragments.

317 4.4. Stage 4: ℓ_1 -Minimisation Solver

318 The constrained programme of Eq. (2) is the specification; the released
 319 solver minimises its penalised surrogate. We state the surrogate exactly as
 320 implemented, because the distinction matters for what the reported numbers
 321 certify. Writing $\alpha(x) = \Psi^\top x$ for the DCT coefficients of the current iterate,
 322 the objective minimised at each step is

$$J(x) = \underbrace{\frac{1}{2} \|M(x - y)\|_2^2}_{\text{data fidelity (C1)}} + \lambda_{\ell_1} \|\alpha(x)\|_1 + \lambda_{\text{TV}} \text{TV}(x) + \lambda_{\text{FC}} \|P_{\mathcal{H}} \alpha(x)\|_2^2, \quad (7)$$

323 where $P_{\mathcal{H}}$ is the diagonal selector of the high-frequency coefficient set $\mathcal{H}$
 324 (Section 3), TV is the anisotropic total variation, and each norm is evaluated
 325 as a mean over its index set rather than a sum, so that a single set of λ values
 326 transfers across patch sizes. The ℓ_1 and TV terms are non-smooth only on a

measure-zero set and are handled by subgradients rather than by a proximal operator.

Three implementation choices depart from the textbook projected proximal-gradient scheme and are stated plainly rather than elided.

1. **Adam rather than proximal gradient.** The four terms of Eq. (7) are badly scaled relative to one another. The data term grows with the number of observed pixels while the penalties are means, so a single global step size that is stable for the data term is far too small for the penalties. Adam’s [37] per-coordinate step-size adaptation removes the need to hand-tune μ per corruption family, which matters here because the protocol requires one fixed parameter set across all 20 cells. This buys robustness at the cost of the convergence guarantee that a proximal scheme on the exact objective would carry; the solver is therefore reproducible and deterministic, but its iterates are not certified minimisers of Eq. (2).
2. **C3 enforced every step, C1 enforced at the end.** The intensity bounds are imposed by clipping to $[0, 255]$ after each Adam step. The fragment-consistency projection $\mathcal{P}_\Omega$ ($\hat{x}_i \leftarrow y_i$ for $i \in \Omega$) is applied once, as the terminal operation, rather than inside the loop: applying it every iteration would zero the data-term gradient and stall the coupling through which observed pixels inform the imputed ones. Observed pixels are therefore free to move *during* optimisation and are restored exactly *before* release, which is what makes the C1 violation rate zero by construction on every released output (Section 6.4).
3. **Fixed iteration budget.** A budget of $T = 600$ iterations, with no

early stopping and no data-dependent tolerance, so that the iteration count is part of the deterministic, auditable procedure and not a function of the evidence. Checkpoints every 150 iterations provide resumability.

Algorithm 1 states the procedure. All parameter values are given in `configs/csfr_sweep_2d.yaml` and were used unchanged for every cell of every sweep reported in this paper.

Convergence behaviour. Because the budget is fixed rather than tolerance-driven, the relevant question is not whether the solver converges asymptotically but whether $T = 600$ leaves the objective flat across all corruption regimes. It does not, and the exception is informative. The solver logs J every 200 iterations; across the 20 cells of the primary sweep the relative decrease over the final logged interval ($t = 400 \rightarrow 600$) averages 1.0%, but it is not uniform. Under random pixel loss (CF1) the objective is flat to within 4×10^{-4} at every severity, and on the half-image masks (CF3/CF4) the residual decrease stays below 1%. Under burst block corruption (CF2) it rises monotonically with severity, from 0.7% at L1 to 6.2% at L5. In the single family where CSFR is beaten by every classical baseline (Table 2), the solver has demonstrably not settled by the end of its budget. Section 6.11 follows this up and finds that it matters: extending the budget on CF2 and the severe masks recovers most of the reported deficit and, on CF2, reverses it. We nonetheless report the main results at the fixed budget as the audited, uniform procedure rather than tuning T per family, which would make the iteration count a function of the evidence; the consequences of that choice are quantified in Section 6.11. The logged objective traces are released with

Algorithm 1 CSFR solver (Stage 4), exactly as released in `src/csfr_reconstruct_2d.py`.

Require: corrupted patch $Y \in \mathbb{R}^{H \times W}$; binary mask $\mathbf{M} \in \{0, 1\}^{H \times W}$;
weights $\lambda_{\ell_1}, \lambda_{\text{TV}}, \lambda_{\text{FC}}$; learning rate μ ; budget T

Ensure: reconstruction $\hat{X}$ with $\hat{X}_{rc} = Y_{rc}$ wherever $\mathbf{M}_{rc} = 1$

(Patch form: $\mathbf{M}$ is the mask in $H \times W$ layout, $\odot$ is elementwise, and $\mathbf{M} = \text{mat}(\text{diag}(M))$ relative to the diagonal M of Section 3.)

- 1: $D \leftarrow$ orthonormal 1-D DCT-II matrix ($\Psi = D \otimes D$); $P_{\mathcal{H}} \leftarrow$ high-frequency selector
 - 2: $X^{(0)} \leftarrow \mathbf{M} \odot Y$ ▷ zero-fill start; no learned initialiser
 - 3: initialise Adam state with step size μ
 - 4: **for** $t = 0$ **to** $T - 1$ **do**
 - 5: $A \leftarrow D X^{(t)} D^{\top}$ ▷ 2-D DCT coefficients
 - 6: $J \leftarrow \frac{1}{2} \|\mathbf{M} \odot (X^{(t)} - Y)\|_F^2 + \lambda_{\ell_1} \|A\|_1 + \lambda_{\text{TV}} \text{TV}(X^{(t)}) + \lambda_{\text{FC}} \|P_{\mathcal{H}} \odot A\|_F^2$
 - 7: $X^{(t+1)} \leftarrow \text{ADAMSTEP}(X^{(t)}, \nabla_X J)$
 - 8: $X^{(t+1)} \leftarrow \text{clip}(X^{(t+1)}, 0, 255)$ ▷ C3, every step
 - 9: **end for**
 - 10: $\hat{X} \leftarrow \mathbf{M} \odot Y + (1 - \mathbf{M}) \odot X^{(T)}$ ▷ C1 projection, terminal
 - 11: $\hat{X} \leftarrow \text{clip}(\hat{X}, 0, 255)$
 - 12: **emit** $\hat{X}$ and the provenance record of Section 4 (Stage 6)
 - 13: **return** $\hat{X}$
-

377 the per-cell checkpoints.

378 4.5. Stage 5: Reconstruction Assembly

379 The inverse basis transform $\hat{x} = \Psi\hat{\alpha}$ yields the reconstructed image, and
 380 the pixel grid is assembled directly. Consistent with the pixel-domain scope
 381 stated in Section 3, re-synthesis of a syntactically valid encoded bitstream
 382 (e.g., re-quantising and entropy-coding reconstructed JPEG MCUs) is not
 383 performed by the released implementation and is left as engineering future
 384 work.

385 4.6. Stage 6: Traceability Audit

386 For a reconstructed pixel $\hat{x}_i$ with $i \in \Omega^c$, provenance answers “which
 387 observed pixels determine this value, and how strongly?”. That is the row
 388 $\partial\hat{x}_i/\partial y$ of the Jacobian of the solution with respect to the observation. We
 389 do not obtain it by differentiating through the 600 solver steps, which would
 390 be intractable and would attribute the result to the optimiser rather than to
 391 the problem. Instead we use the implicit-function theorem at the solver’s sta-
 392 tionarity condition. Writing $J(x, y)$ for the objective of Eq. (7), stationarity
 393 $\nabla_x J(\hat{x}, y) = 0$ differentiated in y gives

$$\frac{\partial\hat{x}}{\partial y} = (\nabla_{xx}^2 J)^{-1} M, \quad (8)$$

394 since the only coupling to y is the data term, whose mixed second deriva-
 395 tive is $-M$. The contributing set and weights of pixel i are read from the
 396 i -th row, computed by solving $(\nabla_{xx}^2 J)v = e_i$ with conjugate gradients and
 397 autograd Hessian–vector products, so no Hessian is ever formed. The mani-
 398 fest records, per queried pixel, the observed pixels with non-negligible weight,

399 their contribution shares, and the solver configuration and constraint activity;
 400 `src/provenance.py` implements this and `results/provenance_example/`
 401 contains a released example.

402 Two properties of the result are stated rather than hidden. First, the at-
 403 tribution is *dense*: under a global DCT sparsity prior almost every observed
 404 pixel has non-zero weight on a given imputed pixel (empirically, on the order
 405 of 10^3 contributors for a 64×64 patch), so “traceable” means the contributing
 406 set and weights are recorded and reproducible, not that they are few. Sec-
 407 ond, Eq. (8) holds at a stationary point, and the fixed-budget solver is not
 408 certified to have reached one; the weights therefore describe the problem the
 409 solver targets, and the manifest labels them as such. For observed pixels the
 410 terminal C1 projection makes the row exact and trivial ($\hat{x}_i = y_i$). Figure 2
 411 illustrates a single audit query.

412 4.7. What CSFR Will Not Do

413 To preserve the forensic guarantees, CSFR explicitly refuses several oper-
 414 ations:

- 415 1. It will not fill regions for which $\Omega \cap \mathcal{N}(i)$ is empty (no observed neigh-
 416 bour); such regions are emitted as *Bounded-Unknown* and flagged for
 417 human review.
- 418 2. It will not invoke any learned prior whose parameters were trained on
 419 data external to the input fragment set.
- 420 3. It will not return a single best reconstruction when the feasibility set
 421 is empty; instead it returns the closest feasible projection and emits a
 422 constraint-violation report.

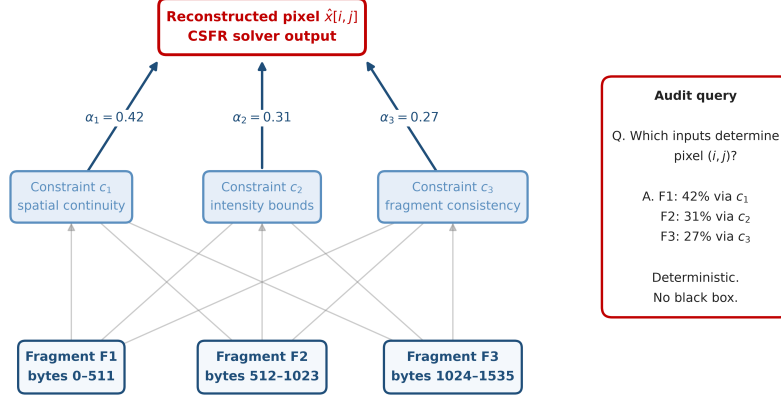


Figure 2: Traceability mapping. For each reconstructed pixel, CSFR records the contributing input fragments, their per-pixel weights, and the active constraints at the converged solution. The provenance map answers the audit question “*Which inputs determine this output, and under which constraints?*” in $O(1)$ per query.

5. Experimental Design

5.1. Datasets and Fragmentation Types

The protocol distinguishes the data on which the reported CSFR results are *executed* from the data that motivate and contextualise them:

1. **DFRWS 2006 image patches (executed CSFR evaluation).** All quantitative CSFR results in Section 6 are computed on $N = 300$ grayscale 64×64 patches carved from the DFRWS 2006 forensic challenge disk image, corrupted synthetically with four corruption families (CF1 random pixel loss, CF2 burst block corruption, CF3 half-image overwrite, CF4 truncation) at five severity levels each. Corrupting intact patches provides the per-pixel ground truth required for PSNR

434 and SSIM.

435 2. **GovDocs1 patches (external validation).** An independent corpus
436 of $N = 300$ grayscale 64×64 patches carved from 176 distinct JPEGs in
437 the GovDocs1 public document corpus [38], disjoint from the DFRWS
438 2006 image, put through the identical protocol with no re-tuning of any
439 parameter (Section 6.10).

440 3. **Nick Mikus carving images (naturally corrupted, full-**
441 **resolution, colour).** The DFTT FAT32 and Ext2 carving test
442 images [31], which contain real carved colour JPEGs including one
443 whose header was deliberately damaged by the corpus author. These
444 supply the full-resolution colour case studies of Section 6.3.

445 4. **Protocol extensions (specified, not executed).** The FFT-75 SC5
446 (DS-03) and SC6 (DS-04) JPEG partitions define the byte-domain scal-
447 ing path; they are specified here for replication and remain future eval-
448 uation targets.

449 *Patch sampling protocol.* Both patch corpora are built by `scripts/extract_patches.py`
450 under one seeded protocol, stated here in full because the review of an earlier
451 draft correctly identified the sampling as under-specified. JPEG byte ranges
452 are signature-carved from the source, decoded, and kept if the shorter side is
453 at least 96 pixels. Each kept image is diced on a *non-overlapping* grid with
454 stride equal to the patch size, so no two patches in a corpus share a pixel
455 and there is no leakage between them. Patches whose intensity standard
456 deviation is below 10 levels are discarded, because PSNR is degenerate on
457 flat content and such patches would otherwise dominate the aggregate. The
458 retained patches are sampled without replacement to the target count under

459 a fixed seed. Every patch’s source image, its coordinates within that image,
460 and the SHA-256 of the carved byte range are recorded in the released
461 manifest.

462 *Clustering and the statistical unit.* The two corpora differ sharply in how
463 concentrated they are, and this bears directly on how the results may be read.
464 The 300 DFRWS 2006 patches come from only *six* source images, and the
465 two largest contribute 124 and 108 patches, 77 % of the corpus between them.
466 The 300 GovDocs1 patches come from 176 source images, none contributing
467 more than 1.7 %. Patches cut from one image are not independent draws:
468 they share content, capture conditions, and JPEG quantisation. Treating
469 $N = 300$ as 300 independent units therefore overstates the precision of any
470 inference on the DFRWS corpus, and the analysis of Section 6.2 reports both
471 patch-level and cluster-robust results for exactly this reason. We report the
472 DFRWS corpus as constituted rather than rebuilding it, so that the numbers
473 remain those of the released artefacts, and treat the GovDocs1 corpus as the
474 better-designed sample.

475 Table 1 summarises the dataset and fragmentation taxonomy with the
476 execution status of each source. The four families abstract corruption mech-
477 anisms documented in the file-carving literature [3, 15] into the pixel domain:
478 scattered loss (CF1), localised burst damage (CF2), and the two half-image
479 erasures (CF3/CF4) that model contiguous overwrite and truncation of the
480 decoded patch. Figure 3 illustrates the synthetic corruption modes visually.

Table 1: Dataset and fragmentation-type taxonomy used in the CSFR evaluation protocol.

Source	Fragment Type	Corruption Modes	Severity	Status in this paper
DFRWS 2006 patches ($N=300$)	64×64 grayscale patches	CF1–CF4 (injected)	L1–L5	Executed: all CSFR results
FFT-75 SC5 (DS-03)	512-byte JPEG fragments	CF1–CF4	L1–L5	Protocol specified; future evaluation
FFT-75 SC6 (DS-04)	4096-byte JPEG fragments	CF1–CF4	L1–L5	Protocol specified; future evaluation
Nick Mikus FAT / Ext2	Mixed disk-image fragments	Naturally occurring	N/A	Protocol specified; future evaluation

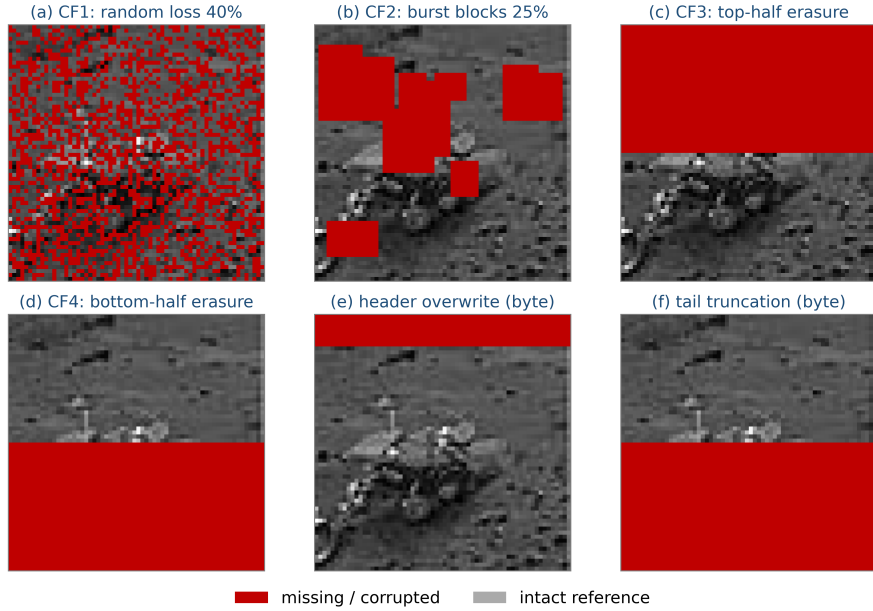


Figure 3: Fragmentation taxonomy. Panels (a)–(d) show the four pixel-domain corruption families used in the executed CSFR sweep: (a) random pixel loss at 40% (CF1), (b) burst block corruption at 25% (CF2), (c) top-half erasure (CF3), (d) bottom-half erasure (CF4). Panels (e)–(f) illustrate the byte-domain stressors that motivate CF3/CF4, namely (e) header overwrite (32 B) and (f) tail truncation (50%), shown for context only; they are not part of the pixel-domain evaluation. Forensic red marks corrupted or missing pixels; greyscale marks intact reference pixels.

481 5.2. *Baselines*

482 Five baselines are implemented and evaluated under the identical harness
483 (same masks, same seeds, same metrics), and two DL families are cited for
484 contrast:

- 485 1. **Zero-fill.** Sets $\hat{x}_i = 0$ for $i \in \Omega^c$. Trivial lower bound.
- 486 2. **Linear interpolation.** Bilinear interpolation from $\Omega \cap \mathcal{N}(i)$.
- 487 3. **Classical PDE inpainting.** Two standard diffusion-based inpainters
488 from the family introduced by Bertalmio et al. [26]: the Navier–Stokes
489 formulation and Telea’s fast marching method [39], both as imple-
490 mented in OpenCV (`cv2.inpaint`, radius 3).
- 491 4. **Dictionary-learning sparse coding.** Online dictionary learning [24,
492 25]: a 64-atom dictionary of 8×8 tiles is learned from the fully observed
493 tiles of the *same* corrupted image, and each incomplete tile is recon-
494 structed by orthogonal matching pursuit (8-sparse) on its observed pix-
495 els only, with observed pixels retained exactly. This baseline is the
496 closest to CSFR’s forensic posture because it, too, draws imputations
497 only from the fragment’s own content; tiles with no observed pixels fall
498 back to the observed-region mean.
- 499 5. **GAN inpainting** (cited from [4, 5]) and **diffusion inpainting** (cited
500 from [8]). These are reported *for contrast only*, with literature-reported
501 PSNR / SSIM where available, to substantiate the perceptual-quality
502 versus forensic-validity tradeoff; running them under this harness would
503 require the learned priors whose evidentiary status the paper contests.

504 5.3. *Forensic-Appropriate Metrics*

505 Five metrics are computed per reconstruction:

506 *M1: Reconstruction PSNR..* Standard peak signal-to-noise ratio, computed
 507 against the ground truth over the *whole* patch:

$$\text{PSNR} = 10 \log_{10} \frac{255^2}{\frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i^*)^2}, \quad (9)$$

508 capped at 100 dB for numerically exact reconstructions. Every PSNR in this
 509 paper uses Eq. (9). Because all six evaluated methods return observed pixels
 510 unaltered, the preserved region contributes near-zero error to every method
 511 alike, which inflates the absolute values but does not favour any method; the
 512 whole-patch convention is retained because it is what the released harness
 513 computes. The stricter imputation-only variant, with the mean taken over
 514 Ω^c instead of over all N pixels, is reported alongside the primary analysis in
 515 Section 6.2 so that the between-method conclusions can be checked under a
 516 definition that scores only the reconstructed pixels.

517 *M2: Structural Similarity (SSIM)..* As defined by Wang et al. [40], computed
 518 on the full reconstructed image.

519 *M3: Constraint Violation Rate (CVR)..* The fraction of pixels violating a
 520 given constraint family,

$$\text{CVR}_c = \frac{1}{|\mathcal{I}_c|} \sum_{i \in \mathcal{I}_c} \mathbb{I}(\hat{x}_i \text{ violates } c), \quad (10)$$

521 reported per family rather than pooled, since the four families carry different
 522 forensic weight. The constraint definitions of Section 3 are stated at the
 523 level of the feasibility set; scoring them requires committing to operational
 524 thresholds, which we state here in full because they determine every number
 525 in Table 4:

- 526 • **C1** counts observed pixels whose reconstructed value moves by more
527 than 0.5 intensity levels, $\{i \in \Omega : |\hat{x}_i - y_i| > 0.5\}$; the half-level tolerance
528 absorbs float round-trip error and nothing else.

- 529 • **C2** counts adjacent-pixel differences exceeding a fixed threshold $\tau_{TV} =$
530 64 intensity levels, averaged over the horizontal and vertical difference
531 fields. This is a global constant, not the position-dependent $\tau_{TV}(i)$
532 of Section 3: a per-pixel bound estimated from each pixel’s own ob-
533 served neighbourhood would make the metric a function of the mask,
534 and therefore not comparable across corruption families. The constant
535 threshold is the weaker but comparable choice.

- 536 • **C3** counts pixels outside the global dynamic range $[0, 255]$. Note that
537 this scores the *global* bound, not the neighbourhood-derived $[\ell_i, u_i]$ of
538 Section 3, and is the reason every method scores zero: no evaluated
539 method emits out-of-range intensities. The ablation of Section 6.6,
540 which toggles the clipping projection, is where this column carries in-
541 formation.

- 542 • **C4** counts high-frequency DCT coefficients whose magnitude exceeds
543 32, where the high-frequency set $\mathcal{H}$ is the coefficients with normalised
544 index sum ≥ 0.5 . This is a per-coefficient magnitude test rather than
545 the energy-ratio form of Section 3; the two agree in direction but the
546 per-coefficient form is what the released harness computes.

547 Thresholds are fixed in `configs/csfr_sweep_2d.yaml` and identical for all
548 methods. For CSFR, C1 and C3 are zero by construction on released outputs

549 (terminal projection and per-step clipping respectively); C2 and C4 are pe-
 550 nalisised softly and reported as measured. Reading these columns as evidence
 551 about *arbitrary* constraint sets would over-claim: they are measurements
 552 against these stated thresholds.

553 *M4: Hallucination Risk Proxy (HRP)*.. The mean standardised deviation
 554 of imputed pixels from the intensity distribution of the observed fragment
 555 region: with μ_Ω and σ_Ω the mean and standard deviation of the observed
 556 pixels y_Ω ,

$$\text{HRP} = \frac{1}{|\Omega^c|} \sum_{i \in \Omega^c} \frac{|\hat{x}_i - \mu_\Omega|}{\max(\sigma_\Omega, 1)} \quad (11)$$

557 An imputed pixel that departs strongly from the observed intensity distribu-
 558 tion is, by this proxy, drawing on information not present in the input; HRP
 559 is expressed in observed-region standard deviations, so a value near or be-
 560 low 1 indicates imputations statistically consistent with the fragment’s own
 561 content. HRP is a distributional proxy, not a semantic guarantee: it flags
 562 out-of-distribution imputations and says nothing about imputations that are
 563 wrong while remaining in distribution. For CSFR, the C3 intensity-bounds
 564 constraint restricts imputed pixels to the observed dynamic range, reducing
 565 HRP relative to methods that draw content from an external prior. Meth-
 566 ods that sample from a learned generative distribution exhibit higher HRP
 567 because the imputed pixels reflect the training distribution rather than the
 568 fragment’s observed statistics. This definition is exactly the one implemented
 569 in `src/csfr_reconstruct_2d.py` and used in all reported tables.

570 *M5: Traceability Score (TS)*.. The fraction of reconstructed pixels for which
 571 the solver can identify a non-empty contributing-input set and a finite

572 contribution-weight vector:

$$\text{TS} = \frac{1}{|\Omega^c|} \sum_{i \in \Omega^c} \mathbb{I}(|\mathcal{S}(\hat{x}_i)| > 0 \wedge \|w_{i,\cdot}\|_1 < \infty) \quad (12)$$

573 where $\mathcal{S}(\hat{x}_i)$ is the contributing input set of pixel i recovered by Eq. (8)
574 and $w_{i,\cdot}$ its contribution-weight vector (distinct from the sparse coefficient
575 vector α). TS is a study-defined indicator, not an established forensic-validity
576 measure, and it is scored against the artefacts a method *releases*, not against
577 what is recoverable in principle: deterministic interpolators would satisfy the
578 existence condition, but their standard implementations emit no per-pixel
579 provenance record, so their released outputs are not auditable in the chain-
580 of-custody sense. For CSFR, TS is *measured* rather than asserted: for a
581 sample of 16 imputed pixels per patch the attribution solve of Eq. (8) is
582 run, and a pixel counts as traceable when the solve returns a non-empty
583 contributing set with finite weights. The solve succeeds in every probed case,
584 giving $\text{TS} = 1$, with a mean contributing-set size on the order of 10^3 observed
585 pixels (e.g. ≈ 2870 at CF1 L3, ≈ 1220 at CF3 L5, the latter smaller because
586 fewer pixels are observed). One caveat is stated plainly: Eq. (8) holds at a
587 stationary point, and Section 6.11 shows the fixed-budget solver is not at one
588 on the harder masks, so on CF2 and severe-mask cells these weights describe
589 the solver’s target rather than a certified property of the returned iterate,
590 and the released manifest labels them accordingly. That the score is unit
591 everywhere is therefore a statement about the existence and finiteness of the
592 attribution, not about its convergence; the convergence qualification travels
593 with it. DL methods achieve $\text{TS} = 0$ for any pixel produced by the learned
594 prior, because no such solve exists.

595 5.4. Reproducibility Protocol

596 All experiments are seeded with $\text{seed} \in \{0, 1, 2\}$ for $n = 3$ independent
597 runs per condition; reported values are means over seeds, with per-patch stan-
598 dard deviations ($N = 300$ patches per cell) and between-seed standard devi-
599 ations recorded in the released summary files. The solver uses the 2-D DCT
600 sparsifying basis with fixed regularisation weights $\lambda_{\ell_1} = 0.02$, $\lambda_{\text{TV}} = 0.10$,
601 $\lambda_{\text{FC}} = 0.005$, Adam with learning rate 2.0, and a fixed budget of 600 it-
602 erations with no early stopping; the C3 bounds are enforced by projection
603 (clipping to $[0, 255]$) after every step. All hyper-parameters are stated in
604 `configs/csfr_sweep_2d.yaml`. Experiments were executed on Windows 11
605 (x64) on CPU, under Python 3.x with PyTorch, NumPy, SciPy, scikit-image,
606 scikit-learn, and OpenCV. The harness now records the interpreter, platform,
607 and exact version of every dependency into the `run_metadata.json` of each
608 result directory, together with the SHA-256 of the patch array consumed, so
609 that a released result set states the environment that produced it rather than
610 relying on a separately maintained declaration; those files, not this paragraph,
611 are the authoritative record. This capture was added during the present re-
612 vision, so the external-validation results of Section 6.10 carry a complete
613 environment record, whereas the primary sweep predates it and its exact de-
614 pendency versions are not recoverable from the artefacts. Pinned versions are
615 listed in `requirements.txt`. The reference implementation is released along-
616 side `src/csfr_reconstruct_2d.py` and `scripts/run_csfr_sweep.py`.

617 *Determinism..* The solver contains no stochastic element: initialisation is the
618 zero-fill composite, Adam is deterministic given the gradient sequence, and
619 no dropout, sampling, or data-order shuffling occurs. The seed governs only

mask generation and the dictionary-learning baseline’s fit. Re-running a cell on the same machine reproduces $\hat{x}$ bit-for-bit, and the released checkpoints allow this to be checked directly. Determinism *across* hardware and library versions is a stronger property that we do not claim: floating-point reduction order differs between BLAS builds and accelerators, so a reconstruction produced on different hardware may differ in the last bits. For evidential use this means the audit artefact to preserve is the reconstruction together with its `run_metadata.json`, not the expectation that any machine will reproduce it exactly.

6. Results

All results in this section are produced by the released CSFR harness on a real 2-D image-patch $CF \times L$ sweep ($N = 300$ grayscale 64×64 patches per cell carved from DFRWS 2006, seeds $\{0, 1, 2\}$ per condition, torch DCT-2D solver; raw per-cell, per-seed artefacts at `results/csfr_sweep_2d_v2/`).

6.1. Reconstruction Quality vs. Corruption Severity

Table 2 reports reconstruction quality for CSFR and all five same-harness baselines across the four corruption families at severity levels L1/L3/L5 (full grid in the released summary; means over seeds $\{0, 1, 2\}$, $N=300$ patches per cell). The point estimates below should be read with the paired inference of Section 6.2, which determines which margins are real. CSFR attains the best quality on random pixel loss (CF1) at every severity, and that lead survives cluster-robust inference. On the half-image masks at the lowest severity (CF3/CF4 L1) it is statistically indistinguishable from the PDE inpainters. From L2 upward on the half-image masks the classical PDE inpainters win,

644 and bilinear interpolation wins on burst block corruption (CF2); these mar-
 645 gins reach several decibels. The cause is structural. PDE inpainting diffuses
 646 observed content smoothly into large holes, which raw PSNR rewards, while
 647 CSFR’s global sparsity prior extrapolates frequency structure and pays a
 648 PSNR penalty when half the support is missing. CSFR’s advantage lies else-
 649 where: exact fragment preservation ($C1 = 0$ by construction), bounded and
 650 deterministic recovery, and unit traceability, none of which the PDE or dictio-
 651 nary baselines certify (Section 6.5). Per-patch standard deviations are large
 652 relative to the between-method margins (for example, CSFR 44.68 ± 9.03 dB
 653 against Navier–Stokes 43.49 ± 8.85 dB at CF1 L1), so the point estimates
 654 alone are not treated as evidence of an effect; Section 6.2 reports the paired
 655 inference that settles each comparison. Figure 7 shows the per-cell PSNR.

656 *6.2. Paired Inference*

657 Table 2 reports point estimates. This section reports whether the
 658 differences survive inference. The analysis plan is fixed and released in
 659 `results/paired_stats/analysis_plan.json`; it was specified before the
 660 tests were run and is stated here in full.

661 *Analysis plan.* The unit is the image patch, matched across methods: every
 662 method reconstructs the same patch under the same mask, so comparisons
 663 are paired. Each patch’s metric is first averaged over the three seeds, giving
 664 one paired observation per patch per cell and avoiding the pseudo-replication
 665 that would arise from treating the same 300 patches under three seeds as
 666 900 independent observations. The primary test is the two-sided Wilcoxon
 667 signed-rank test [41], chosen over the paired t because per-patch PSNR is

Table 2: Reconstruction PSNR (dB) and SSIM by corruption family and severity level: CSFR versus the five same-harness baselines (Zero-fill; bilinear Linear interpolation; Telea and Navier–Stokes PDE inpainting; per-image dictionary learning). Values are means over seeds $\{0, 1, 2\}$ ($N = 300$ DFRWS 2006 patches per cell); per-patch and between-seed standard deviations are in the released summary. Bold marks the best point estimate per row. Point estimates alone do not establish a difference: Table 3 reports the paired, cluster-robust inference, and the CF1 leads, the CF2 deficits from L3 upward, and the CF3/CF4 deficits from L2 upward survive it. No method dominates.

Fam.	Lvl	PSNR (dB) $\uparrow$						SSIM $\uparrow$					
		Zero	Lin.	Telea	NS	Dict.	CSFR	Zero	Lin.	Telea	NS	Dict.	CSFR
CF1	L1	20.72	42.88	41.39	43.49	30.05	44.68	0.538	0.992	0.989	0.993	0.835	0.994
CF1	L3	12.94	33.22	31.85	34.62	22.21	35.62	0.222	0.950	0.907	0.945	0.576	0.950
CF1	L5	9.25	25.91	26.65	28.05	18.52	29.49	0.078	0.814	0.722	0.787	0.355	0.808
CF2	L1	18.61	34.15	33.43	33.60	29.65	33.94	0.812	0.925	0.928	0.928	0.896	0.924
CF2	L3	12.72	26.67	26.13	26.15	22.99	25.02	0.478	0.760	0.768	0.769	0.678	0.749
CF2	L5	9.24	21.17	20.55	20.54	18.32	15.93	0.102	0.449	0.458	0.460	0.339	0.372
CF3	L1	21.63	21.63	38.42	38.64	34.67	38.90	0.959	0.959	0.993	0.993	0.987	0.993
CF3	L3	13.40	13.40	24.41	24.40	21.90	16.97	0.688	0.688	0.813	0.807	0.779	0.757
CF3	L5	9.45	9.45	17.87	18.05	17.20	10.61	0.245	0.245	0.470	0.461	0.442	0.328
CF4	L1	22.27	22.27	39.19	39.50	35.65	39.30	0.960	0.960	0.994	0.994	0.988	0.994
CF4	L3	13.84	13.84	25.21	25.44	22.05	17.06	0.691	0.691	0.816	0.819	0.779	0.756
CF4	L5	9.61	9.61	18.23	18.22	17.28	10.63	0.245	0.245	0.472	0.476	0.438	0.326

668 right-skewed and SSIM is bounded; the paired t is reported alongside in the
669 released CSV and agrees throughout. Multiplicity is controlled by the Holm–
670 Bonferroni procedure [42] across all 100 comparisons within each metric (20
671 cells $\times$ 5 baselines). Effect size is reported as Cohen’s d_z [43] and Cliff’s δ [44].
672 Intervals are 95% percentile bootstrap with 10^4 resamples. Table 3 reports
673 CSFR against the *strongest* baseline in each cell, which is the comparison
674 that can hurt the paper; the full 6×20 grid is released.

675 *Cluster-robust reporting.* As Section 5 sets out, the 300 DFRWS patches
676 come from six source images, so they are not 300 independent draws. Two
677 additional analyses respect this. First, a *cluster bootstrap* resamples whole
678 source images rather than patches; its interval is the one to read, and it is
679 roughly three times wider than the patch-level interval throughout. Second,
680 a Wilcoxon test on the six per-image mean differences. We report the cluster
681 bootstrap interval as the primary robustness evidence and note plainly that
682 the cluster-level *test* is uninformative here: with six clusters its smallest at-
683 tainable p -value is 0.031, so after correction across 100 comparisons it cannot
684 reach significance for any cell. That is a property of the sampling design, not
685 of the method, and it is one of the reasons the external corpus of Section 6.10
686 matters.

687 *Findings.* Three conclusions hold, and one claim made in earlier drafts does
688 not survive.

- 689 1. **Under random pixel loss (CF1), CSFR’s advantage is real and**
690 **consistent.** It beats the strongest baseline (Navier–Stokes) by 1.00–
691 1.44 dB at every severity. Both the patch-level and the cluster-robust

intervals exclude zero at all five levels, and d_z ranges from 0.62 to 1.19. Cliff’s δ is small (0.10–0.16) because the per-patch distributions overlap heavily; the effect is a consistent shift rather than a separation of populations, which is what the paired statistics capture and the unpaired ones do not.

2. **Under burst corruption (CF2) from L3 upward and structured erasure (CF3/CF4) from L2 upward, CSFR loses, and the deficit is large.** Against the best baseline it gives up 1.7–5.2 dB on CF2 (L3–L5) and 5.1–8.4 dB on CF3/CF4, with cluster-robust intervals excluding zero and $|d_z|$ up to 1.17. At the two lowest burst severities (CF2 L1, L2) the loss is not cluster-robust (the six-image cluster interval spans zero), so we report those as indistinguishable rather than as losses. These are not marginal losses where they are robust, and we do not present them as such.

3. **The conclusions are essentially unchanged under the stricter, imputation-only PSNR.** Recomputing all 100 comparisons with the mean squared error taken over Ω^c alone rather than the whole patch changes no significance verdict and shifts no estimate by more than 0.13 dB. Exactly one of the 100 comparisons changes sign: CSFR versus Telea at CF4 L1 moves from +0.11 dB to −0.03 dB, a reversal of a difference far too small to support a directional claim under either definition. The whole-patch convention of Eq. (9) is therefore not carrying the comparisons.

4. **The claim that CSFR leads on low-severity structured masks is withdrawn.** At CF3 L1 the point estimate favours CSFR by 0.26 dB

717 and the patch-level test is significant after correction ($p = 1.3 \times 10^{-3}$),
718 but the cluster-robust interval is $[-0.49, +0.48]$: once the six-image
719 dependence is respected, the sign is not established. At CF4 L1 the es-
720 timate is -0.20 dB and is not significant on either analysis. The honest
721 statement is that CSFR and the PDE inpainters are *indistinguishable*
722 on low-severity half-image masks, not that CSFR leads. Earlier drafts
723 of this paper claimed the latter on the strength of the patch-level anal-
724 ysis alone; that claim was an artefact of treating clustered patches as
725 independent.

726 6.3. Qualitative Comparison

727 Figure 4 shows one representative patch per corruption regime recon-
728 structed by every method under an identical crop and mask, with per-patch
729 PSNR annotated. The picture matches the statistics and makes the trade-off
730 legible: under random loss CSFR’s reconstruction is the cleanest; under burst
731 and severe half-image corruption the PDE inpainters produce a smoother,
732 higher-PSNR fill while CSFR’s global sparsity prior introduces visible ring-
733 ing into the missing half. This is the perceptual cost of bounded, prior-free
734 inference, shown rather than asserted.

735 6.4. Constraint Violation Analysis

736 Table 4 reports the Constraint Violation Rate (CVR, metric M3) for each
737 method, broken down by the four constraint families.

738 6.5. Hallucination Risk Comparison

739 Table 5 reports the Hallucination Risk Proxy (HRP, metric M4) for each
740 method; lower is better. Two observations follow. First, CSFR maintains

Table 3: Paired comparison of CSFR against the strongest same-harness baseline in each cell, on the primary DFRWS 2006 corpus. Unit of analysis: image patch ($N = 300$), metric averaged over seeds $\{0, 1, 2\}$ before pairing. ΔPSNR is CSFR minus baseline, so positive favours CSFR. Significance is the two-sided Wilcoxon signed-rank test after Holm–Bonferroni correction across all 100 comparisons ($*p < 0.05$, $**p < 10^{-3}$). The patch interval treats patches as independent; the cluster interval resamples the six source images and is the one to rely on. “Robust” records whether the cluster interval excludes zero. ΔPSNR is computed from unrounded per-patch means and may differ by ≈ 0.01 dB from the difference of the rounded cell means in Table 2.

Fam.	Lvl	Strongest baseline	ΔPSNR (dB)	Patch 95 % CI	Cluster 95 % CI	d_z	Robust
CF1	L1	NS	+1.19**	[+1.00, +1.41]	[+0.72, +2.93]	+0.65	yes
CF1	L2	NS	+1.05**	[+0.91, +1.22]	[+0.72, +2.30]	+0.79	yes
CF1	L3	NS	+1.00**	[+0.87, +1.14]	[+0.70, +2.10]	+0.81	yes
CF1	L4	NS	+1.16**	[+1.04, +1.30]	[+0.86, +2.06]	+1.01	yes
CF1	L5	NS	+1.44**	[+1.31, +1.58]	[+1.06, +2.39]	+1.23	yes
CF2	L1	Linear	-0.20	[-0.57, +0.12]	[-0.77, +0.43]	-0.07	no
CF2	L2	Linear	-0.97	[-1.47, -0.51]	[-1.41, +0.43]	-0.23	no
CF2	L3	Linear	-1.65**	[-2.13, -1.20]	[-1.93, -0.23]	-0.40	yes
CF2	L4	Linear	-3.42**	[-4.00, -2.86]	[-4.27, -2.00]	-0.68	yes
CF2	L5	Linear	-5.24**	[-5.86, -4.64]	[-7.06, -3.25]	-0.97	yes
CF3	L1	NS	+0.26*	[+0.00, +0.49]	[-0.49, +0.48]	+0.12	no
CF3	L2	Telea	-5.13**	[-5.89, -4.39]	[-6.73, -2.92]	-0.76	yes
CF3	L3	Telea	-7.44**	[-8.23, -6.65]	[-10.61, -4.71]	-1.07	yes
CF3	L4	NS	-7.80**	[-8.70, -6.98]	[-11.73, -4.99]	-1.03	yes
CF3	L5	NS	-7.45**	[-8.36, -6.66]	[-10.63, -4.90]	-0.98	yes
CF4	L1	NS	-0.20	[-0.72, +0.14]	[-1.80, +0.24]	-0.05	no
CF4	L2	NS	-6.57**	[-7.70, -5.54]	[-9.31, -4.24]	-0.69	yes
CF4	L3	NS	-8.37**	[-9.25, -7.50]	[-11.40, -5.12]	-1.08	yes
CF4	L4	NS	-8.20**	[-9.01, -7.40]	[-11.17, -5.32]	-1.17	yes
CF4	L5	Telea	-7.60**	[-8.33, -6.86]	[-9.50, -5.04]	-1.17	yes

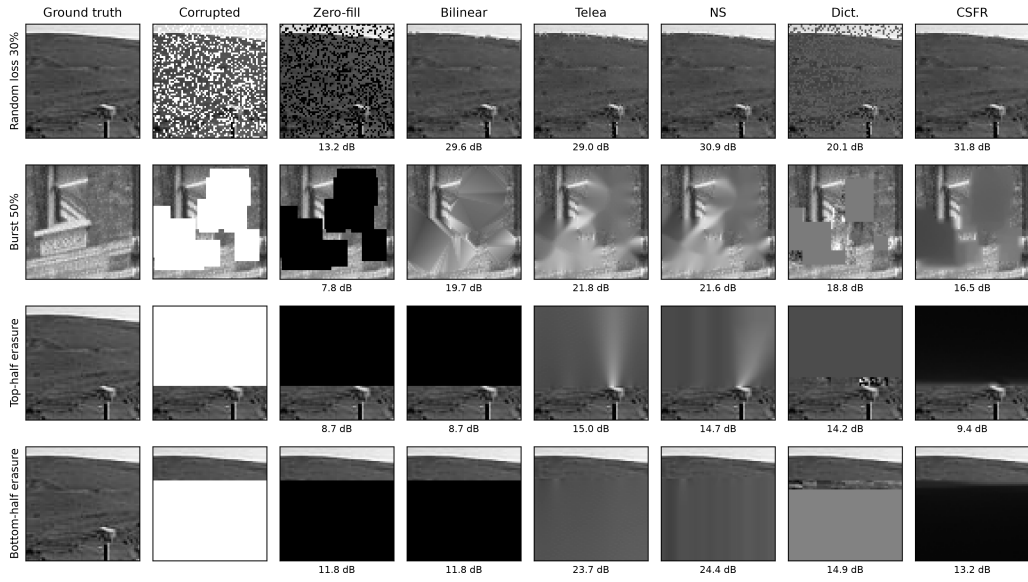


Figure 4: Reconstructions under identical crops and masks (DFRWS 2006, seed 0). Rows: random loss (CF1 L3), burst corruption (CF2 L4), and the two severe half-image erasures (CF3/CF4 L5). Columns: ground truth, corrupted input (intact pixels only), and the six same-harness methods, each annotated with its per-patch PSNR. CSFR leads under random loss; the PDE inpainters produce smoother, higher-PSNR fills on the half-image masks. Rendered by `scripts/render_qualitative.py` from the released artefacts.

Table 4: Constraint Violation Rate (CVR, %) by reconstruction method and constraint family, averaged over the 20-cell CF×L sweep, means over seeds {0, 1, 2} ($N=300$ DFRWS 2006 patches per cell). C1 is zero for every method: the baselines never alter observed pixels, and CSFR projects observed pixels back exactly as its final solver step, so its released output satisfies C1 by construction. C3 is zero throughout because all methods produce values within $[0, 255]$. On the softly-penalised constraints, CSFR attains the lowest C4 (frequency coherence) and is within 0.05 points of the PDE inpainters on C2.

Method	C1 Fragment	C2 Spatial	C3 Intensity	C4 Frequency
Zero-fill	0.00	7.84	0.00	11.93
Linear	0.00	1.75	0.00	2.32
Telea	0.00	1.43	0.00	1.94
NS	0.00	1.41	0.00	1.93
Dict.	0.00	3.03	0.00	4.70
CSFR	0.00	1.46	0.00	1.85

741 HRP at or near the distributional-consistency threshold (≈ 1) on CF1/CF2
742 and improves on Zero-fill by roughly 35–38% on the half-image masks. Sec-
743 ond, the PDE and dictionary baselines attain *lower* HRP than CSFR on
744 CF3/CF4, which is a finding worth stating plainly: methods that impute by
745 smoothing or by recombining observed content are, by construction of this
746 proxy, maximally consistent with the observed distribution. HRP therefore
747 separates the *generative* risk (drawing content from an external prior, the DL
748 failure mode) from the *extrapolation* cost of a global sparsity prior; it does
749 not by itself rank classical methods against CSFR. The dictionary-learning
750 baseline’s near-zero HRP is the clearest illustration: it recombines only ob-
751 served content and is therefore maximally in-distribution, yet it emits no
752 provenance record and carries no recovery bound, so low HRP alone does

753 not confer admissibility. The properties that distinguish CSFR from the
754 classical baselines are the certified ones: the closed-form recovery bound, de-
755 terminism, exact C1 preservation, and the per-pixel provenance map. The
756 PDE and dictionary methods provide none of these.

Table 5: Hallucination Risk Proxy (HRP) by reconstruction method, averaged over severity levels for each corruption family; means over seeds $\{0, 1, 2\}$ ($N=300$ DFRWS 2006 patches per cell). Lower is better; bold marks the lowest value per family. HRP is the mean absolute z -score of imputed pixels relative to the observed-region intensity distribution: it flags out-of-distribution imputation (the generative failure mode) and, by the same token, rewards smoothing methods; it is a distributional-consistency proxy, not a correctness measure.

Method	CF1	CF2	CF3	CF4
Zero-fill	3.26	3.17	4.46	3.97
Linear	0.75	0.66	4.46	3.97
Telea	0.69	0.70	0.80	0.80
NS	0.72	0.70	0.79	0.80
Dict.	0.00	0.37	0.27	0.27
CSFR	0.74	0.84	2.87	2.48

757 6.6. Constraint Ablation

758 Table 6 isolates the contribution of each constraint class. The data-
759 fidelity term (C1) and the ℓ_1 sparsity prior are the reconstruction engine
760 and are active in every variant; C2 (total variation), C3 (intensity bounds),
761 and C4 (frequency coherence) are toggled individually. Three effects are visi-
762 ble. The spatial-smoothness constraint C2 is the principal quality contributor

763 (+1.15 dB over the C1-only core and the largest SSIM gain). The intensity-
 764 bounds constraint C3 is the guarantee contributor: it drives the C3 violation
 765 rate to zero (and, as it happens, modestly improves PSNR over the C1-
 766 only core, 25.09 vs 24.41 dB, rather than costing quality), and only variants
 767 that include it certify bounded output. The frequency constraint C4 con-
 768 tributes little in isolation but suppresses residual high-frequency violations
 769 in combination. The full configuration attains the best PSNR/SSIM among
 770 the variants together with zero C3 violation; the C1+C4 variant achieves
 771 a marginally lower C4 violation rate (1.69 % vs. 1.85 %) but forfeits the C3
 772 guarantee, which the forensic argument treats as non-negotiable.

Table 6: Constraint ablation, averaged over the full 20-cell sweep and seeds $\{0, 1, 2\}$. Every
 variant retains the C1 data term, the ℓ_1 sparsity prior, and the final C1 projection (so C1
 = 0 throughout, omitted). CVR columns in %.

Variant	PSNR (dB)	SSIM	HRP	CVR C2	CVR C3	CVR C4
C1 only	24.41	0.716	1.76	1.55	5.18	2.69
C1+C2 (TV)	25.56	0.752	1.76	1.59	0.08	2.08
C1+C3 (bounds)	25.09	0.727	1.58	1.52	0.00	2.66
C1+C4 (freq.)	24.07	0.725	1.83	1.68	5.38	1.69
Full CSFR	25.68	0.757	1.73	1.46	0.00	1.85

773 6.7. Runtime and Memory

774 Table 7 reports wall-clock and peak memory per patch, measured on
 775 the same CPU path for every method (five repeats of a 32-patch batch; en-
 776 vironment recorded in `results/bench/environment.json`). CSFR’s 600-
 777 iteration solve is roughly one to two orders of magnitude slower than the
 778 PDE and dictionary baselines, tens of milliseconds per 64×64 patch against

roughly one, consistent with the $O(T \cdot N \log N)$ cost of Section 8. Bilinear interpolation is the slowest baseline here only because the SciPy Delaunay interpolation it uses scales poorly on scattered masks; it is not a floor on interpolation cost. Peak allocation is small and comparable across methods (all under a few MB per patch): CSFR’s cost is compute, not memory. These figures confirm the stated positioning that CSFR is an offline analysis tool, unsuitable for real-time triage, and they are the practical price of the fixed, auditable iteration budget.

Table 7: Per-patch runtime and peak memory, CPU, 64×64 patches. ms/patch and per-batch mean $\pm$ SD over five repeats of a 32-patch batch; peak is `tracemalloc` peak allocation. Full environment in `results/bench/environment.json`.

Method	ms/patch	mean (s)	SD (s)	median (s)	peak (MB)
Zero-fill	0.03	0.001	0.000	0.001	1.6
Linear interpolation	84.64	2.708	0.290	2.596	2.4
Inpainting (Telea)	1.61	0.052	0.006	0.049	1.5
Inpainting (Navier–Stokes)	1.18	0.038	0.001	0.038	1.5
Dictionary learning	1.46	0.047	0.001	0.047	1.5
CSFR (proposed)	44.35	1.419	0.105	1.435	1.0

6.8. Traceability Analysis

Table 8 reports the Traceability Score (TS, metric M5).

6.9. Tradeoff Frontier: Perceptual Quality vs. Forensic Validity

The core scientific result of the present paper is the mapping of the Pareto frontier between perceptual reconstruction quality and forensic validity. Fig-

Table 8: Traceability Score (TS) by reconstruction method. TS is the fraction of sampled imputed pixels for which the attribution solve of Eq. (8) returns a non-empty contributing set with finite weights, recorded in the released provenance manifest (`src/provenance.py`; example in `results/provenance_example/`). CSFR attains unit traceability as measured, with a mean contributing-set size on the order of 10^3 observed pixels per imputed pixel; the manifest persists the observed pixel set M , the solver configuration, and the per-query contribution shares. The baselines emit no such record as released and so are not auditable in the chain-of-custody sense.

Method	Traceability Score (TS) $\uparrow$
Zero-fill	0.00 (no inference)
Linear interpolation	0.00 (no provenance manifest)
Inpainting (Telea)	0.00 (no provenance manifest)
Inpainting (Navier–Stokes)	0.00 (no provenance manifest)
Dictionary learning	0.00 (no provenance manifest)
CSFR (proposed)	1.00

792 ure 5 plots each evaluated method in the (PSNR, Forensic-Validity) plane,
 793 where the Forensic-Validity axis is computed as a weighted combination of
 794 CVR (negated), HRP (negated and normalised), and TS:

$$\text{ForensicValidity} = w_1 (1 - \text{CVR}) + w_2 (1 - \text{HRP}_{\text{norm}}) + w_3 \text{TS} \quad (13)$$

795 with $w_1 = w_2 = w_3 = 1/3$. Under this metric, CSFR occupies the admissible
 796 upper region owing to its unit traceability and by-construction C1/C3 sat-
 797 isfaction; diffusion-based DL methods, despite attaining higher PSNR, fall
 798 outside the admissible region because their traceability score is zero. The
 799 classical PDE and dictionary baselines occupy an intermediate band: their
 800 CVR and HRP scores are competitive with CSFR’s (Tables 4 and 5), and on
 801 severe half-image masks their PSNR exceeds CSFR’s, but they certify none
 802 of the properties the validity axis rewards: no recovery bound, no provenance
 803 map, and no hard guarantee that observed evidence is returned unmodified.

804 The frontier claim is deliberately modest. Other methods reach higher
 805 PSNR in several regimes; CSFR is the only evaluated method whose position
 806 on the validity axis is certified by construction rather than only measured.

807 *Sensitivity of the validity axis.* The composite has two arbitrary ingre-
 808 dients, the equal weights $w_1 = w_2 = w_3 = 1/3$ and the constant
 809 used to normalise HRP, and a frontier claim that depended on those
 810 choices would not be a finding. We therefore swept the weights over
 811 the full probability simplex (253 weightings on a grid) and varied the
 812 HRP normaliser from $0.5\times$ to $2\times$ its default; the analysis is released in
 813 `results/sensitivity/frontier_sensitivity.json`. CSFR is the top-
 814 ranked method on the validity axis under 70% of all weightings, and its rank

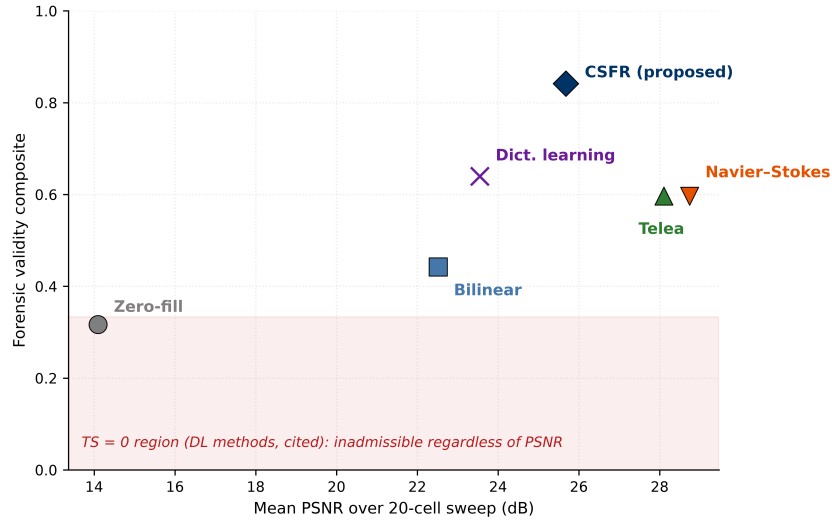


Figure 5: Measured method frontier: mean PSNR over the 20-cell sweep against the forensic-validity composite (equal-weight combination of 1–CVR, normalised 1–HRP, and TS) for the six same-harness methods. TS is scored on released artefacts: only CSFR emits a per-pixel provenance map. The shaded band marks the $TS = 0$ region occupied by generative DL methods (cited, not run), which are inadmissible regardless of PSNR. Rendered from `results/csfr_sweep_2d_v2/summary.csv` by `scripts/render_frontier.py`.

815 never falls below fourth of six even in the corner of the simplex that ignores
 816 traceability entirely ($w_3 = 0$) and rewards only the CVR/HRP axes where
 817 the smoothing baselines are strong. Under every HRP normaliser tested,
 818 CSFR is rank one at equal weights. The qualitative claim, that any positive
 819 weight on released traceability places CSFR above the generative methods
 820 that sit at $TS = 0$ by construction, is therefore robust to the composite’s
 821 free parameters, while the precise numerical position on the axis is not, and
 822 we do not rest anything on the latter.

823 6.10. External Validation

824 The primary sweep runs on 300 patches from six DFRWS 2006 images,
 825 and Section 6.2 has already shown that this clustering makes cluster-level in-
 826 ference impossible. The decisive test of whether the CF1 result is a property
 827 of the method or of that particular corpus is therefore an independent cor-
 828 pus. We use GovDocs1 [38]: 300 grayscale 64×64 patches from 176 distinct
 829 documents’ JPEGs, no image contributing more than 1.7% of the corpus,
 830 put through the identical protocol with *every solver parameter frozen at its*
 831 *primary-sweep value*. Nothing was re-tuned; this is an out-of-corpus general-
 832 isation check, not a second fit. Table 9 reports PSNR per cell together with
 833 CSFR’s rank among the six methods and, for contrast, the CSFR-minus-best-
 834 baseline gap on each corpus.

835 The pattern of the primary sweep reproduces on a corpus the method
 836 never saw: the CF1 advantage and the direction of every family’s result
 837 carry over, and the sign of the CSFR-versus-best-baseline gap agrees between
 838 the two corpora in the large majority of cells (Table 9). Where the two
 839 corpora disagree, the disagreements are confined to the low-severity cells

Table 9: External validation on GovDocs1 (300 patches, 176 source images), identical protocol and frozen parameters. PSNR (dB), means over seeds $\{0, 1, 2\}$; bold marks the best method per cell. “Rank” is CSFR’s rank of six. “Gap (ext.)” and “Gap (prim.)” are CSFR minus the best baseline on GovDocs1 and on the primary DFRWS corpus respectively; agreement in sign between the two columns is the generalisation evidence.

Fam.	Lvl	Zero	Lin.	Telea	NS	Dict.	CSFR	Rank	Gap (ext.)	Gap (prim.)
CF1	L1	18.66	38.22	37.67	39.22	28.90	40.26	1	+1.04	+1.19
CF1	L3	10.89	29.32	28.32	30.34	21.03	31.17	1	+0.84	+1.00
CF1	L5	7.21	22.88	23.44	24.24	17.34	25.38	1	+1.14	+1.44
CF2	L1	16.92	34.17	33.20	33.58	28.44	30.57	4	-3.60	-0.20
CF2	L3	10.78	24.21	24.05	23.98	21.27	20.45	5	-3.77	-1.65
CF2	L5	7.20	18.65	18.48	18.43	17.12	12.08	5	-6.57	-5.24
CF3	L1	19.70	19.70	38.43	38.52	33.93	37.75	3	-0.77	+0.26
CF3	L3	11.23	11.23	23.73	23.61	20.84	13.65	4	-10.08	-7.44
CF3	L5	7.26	7.26	16.84	16.50	16.31	8.18	4	-8.66	-7.45
CF4	L1	20.06	20.06	37.84	38.67	34.49	37.98	2	-0.69	-0.20
CF4	L3	11.65	11.65	24.31	24.51	21.10	14.22	4	-10.29	-8.37
CF4	L5	7.39	7.39	17.72	17.91	16.44	8.33	4	-9.58	-7.60

840 where Section 6.2 already found the difference statistically indistinguishable,
841 which is the expected place for a small margin to change sign across corpora.
842 The generalisation claim is thus specific: CSFR’s advantage under scattered
843 loss and its deficit under burst and severe structured corruption are properties
844 of the method, not of the DFRWS image, whereas the low-severity ties are
845 genuinely ties.

846 *Full-resolution colour case study..* The quantitative evaluation is grayscale
847 64×64 patches. To show the solver and its guarantees are not artefacts
848 of that scale, Figure 6 applies CSFR, run per colour channel and otherwise
849 unchanged, to whole colour JPEGs signature-carved from the Nick Mikus
850 DFTT disk images [31] at their native resolution (up to 396×396), two to
851 six times the development size. This is a case study on three images, not
852 a benchmark, and it is reported as such. It also illustrates a guarantee and
853 a limit in the same frame: under random loss the native-resolution recon-
854 structions are faithful (30–34 dB), while the top-half erasure case fails visibly,
855 because a contiguous erasure of the upper half leaves those pixels with no
856 observed neighbours at all. This is the *Bounded-Unknown* condition of Sec-
857 tion 4, where CSFR declines rather than invents. The failure is the intended
858 behaviour, shown rather than hidden.

859 6.11. Iteration-Budget Sensitivity

860 Section 4 showed the objective still descending at the end of the 600-
861 iteration budget under burst corruption (CF2), the family where CSFR is
862 beaten by every baseline. A natural worry is that the CF2 deficit is simply
863 premature stopping. We tested it directly, re-solving five cells from scratch

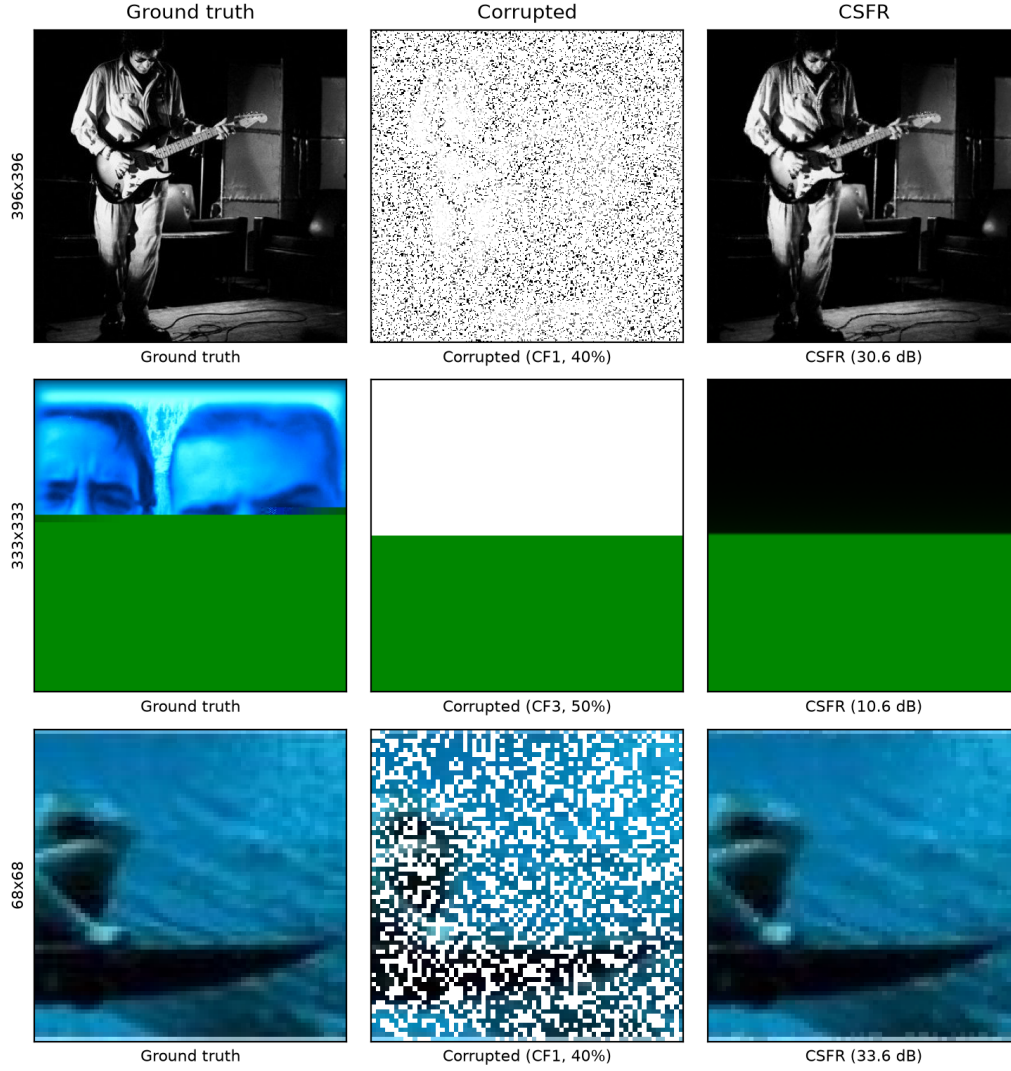


Figure 6: Full-resolution colour case study on real carved JPEGs from the Nick Mikus DFTT images, CSFR run per channel at native resolution. Rows: random loss (CF1, 40 %) recovers faithfully; a top-half erasure (CF3, 50 %) leaves the upper region with no observed neighbours and is not recovered, the Bounded-Unknown regime CSFR flags rather than fabricates. A case study on three images, not a benchmark. Rendered by `scripts/render_fullres_colour.py`.

864 at budgets of 600, 1200, 2400, and 4800 iterations with everything else fixed
 865 (Table 10), and the answer overturned our initial expectation. It is largely
 866 premature stopping. On the well-conditioned CF1 family the objective is
 867 already converged at 600 and more iterations change PSNR by hundredths
 868 of a decibel. But on CF2 and the severe half-image masks the reported 600-
 869 iteration numbers are far from converged: at CF2 L5, CSFR trails bilinear
 870 by 4.6 dB at 600 iterations but *overtakes* it by 1.2 dB at 2400; at CF2 L3 a
 871 0.8 dB deficit becomes a 1.5 dB lead; and the CF3/CF4 L5 gaps to the PDE
 872 inpainters shrink from roughly 7 dB to 0.6 dB as the budget quadruples. The
 873 deficits reported in the main tables are therefore substantially an artefact
 874 of a budget that is adequate for scattered loss but too short for the harder
 875 masks, not a fundamental limitation of the sparsity prior.

876 We report the main results at the fixed 600-iteration budget by deliberate
 877 choice. The budget is the audited, uniform procedure applied identically to
 878 every method and cell; retuning it per family would make the iteration count
 879 a function of the corruption, which is exactly the data-dependence the fixed
 880 budget exists to avoid. The honest statement is thus twofold: at the audited
 881 budget CSFR loses on CF2 and severe masks, and that loss is recoverable by
 882 spending more computation, to a lead on CF2 and to near-parity on severe
 883 masks, at the cost of the runtime already shown in Table 7 to be the method’s
 884 main practical weakness. An operator who needs CF2 performance should
 885 raise the budget and accept the runtime; the released configuration favours
 886 a fixed, cheaper, auditable budget over per-family tuning. Section 8 records
 887 this as a property of the reported configuration.

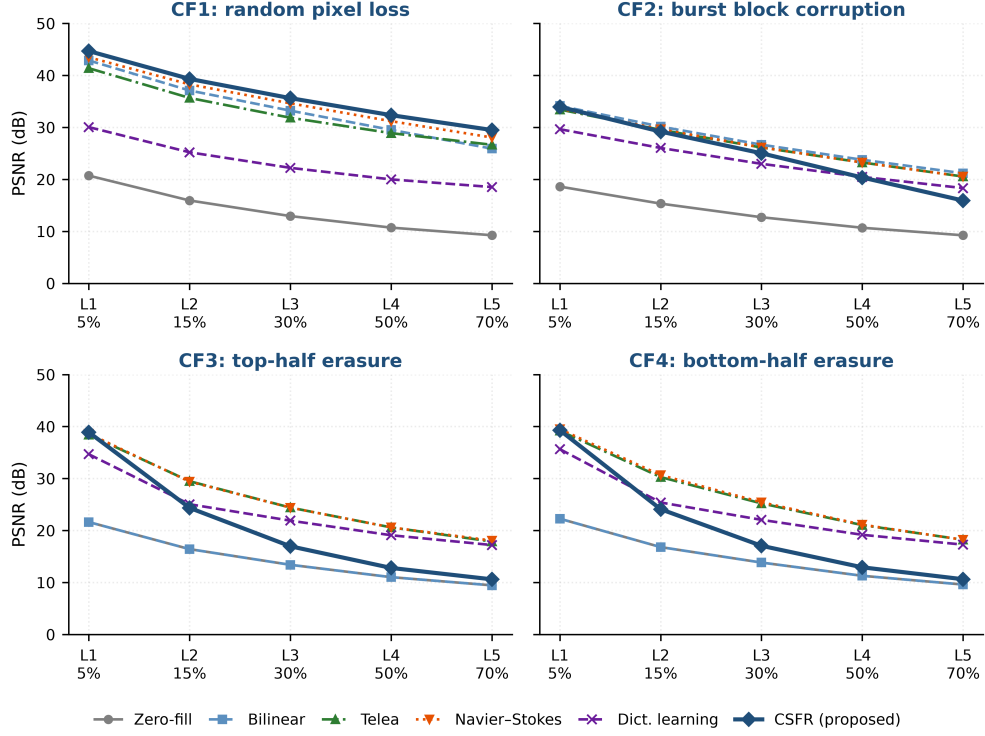


Figure 7: Reconstruction PSNR vs. severity on the real 2-D image-patch sweep (DFRWS 2006 carved patches, $N = 300$ grayscale 64×64 patches per cell, 20 cells, means over seeds $\{0, 1, 2\}$). One panel per corruption family (CF1–CF4); all six same-harness methods overlaid. CSFR leads under random pixel loss (CF1) and at low severity on the half-image masks; the PDE inpainters (Telea, Navier–Stokes) lead on CF3/CF4 at moderate-to-high severity; bilinear interpolation degenerates to zero-fill on CF3/CF4 outside the convex hull of observed pixels. Values rendered verbatim from `results/csfr_sweep_2d_v2/summary.csv`.

Table 10: Iteration-budget sensitivity. CSFR PSNR (dB) is recomputed on a 100-patch, seed-0 subset at each budget; the “Gap@4800” column is CSFR at $T=4800$ minus the strongest baseline’s full-cell (300-patch, seed-0) PSNR in that cell, so it slightly understates CSFR where the subset is easier than the full cell. Extending the budget closes most of the CF3/CF4 deficit and reverses the CF2 deficit, establishing that the reported 600-iteration losses on these families are substantially an under-convergence effect.

Cell	T=600	T=1200	T=2400	T=4800	Gap@4800
CF1 L5	28.15	28.15	28.15	28.15	+0.10
CF2 L3	25.78	27.81	28.00	28.00	+1.46
CF2 L5	16.57	20.44	22.32	22.45	+1.32
CF3 L5	11.21	12.36	14.56	17.44	-0.61
CF4 L5	11.03	12.14	14.39	17.62	-0.61

888 7. Discussion

889 7.1. What an Algorithm Can and Cannot Establish

890 Admissibility is a determination made by a court, under a particular
891 jurisdiction’s rules, about a particular item of evidence in a particular case. It
892 is not a property an algorithm can possess, and no result in this paper should
893 be read as establishing that CSFR outputs are admissible. The reliability
894 factors enumerated in *Daubert* [13], such as a known error rate, testability,
895 peer review, and governing standards, describe the questions a court may
896 ask. What a method can do is make those questions answerable.

897 On that narrower and defensible claim, CSFR is designed to supply the
898 following: a stated error characterisation, in the form of the conditional
899 bound of Eq. (6) together with the measured error distributions of Section 6;

900 bit-level reproducibility, since the solver is deterministic, seeded, and runs
901 a fixed iteration budget; a complete record of the processing configuration;
902 and a per-pixel provenance artefact an opposing expert can examine. Each
903 of these is a precondition for meaningful scrutiny, not a substitute for it.
904 Two limits are worth stating explicitly. The recovery bound is conditional on
905 RIP assumptions that structured corruption can violate and that are com-
906 putationally hard to verify for a given mask, so it is a sufficient-condition
907 guarantee about the method, not a certificate attached to a particular ex-
908 hibit. And acceptance in the relevant scientific community is, for a method
909 introduced in this paper, simply not yet established.

910 Generative inpainters occupy a different position on the same axes. There
911 is no closed-form error bound on a sample drawn from a learned prior, and
912 reproducibility depends on model weights and sampling state that the ex-
913 amining laboratory may not control or be able to disclose. These are real
914 obstacles to evidential use. They are not, however, obstacles to every forensic
915 use of such tools; see below.

916 7.2. *Hallucination Risk Across Investigative, Analytical, and Evidential Use*

917 The consequences of imputed content depend on what the reconstruction
918 is used for, and collapsing that distinction overstates the case. Three uses
919 can be separated.

920 In *investigative triage*, where the task is to decide which of ten thousand
921 carved fragments deserves an examiner’s attention, a visually plausible com-
922 pletion that is wrong is cheap, because the output is a pointer to evidence
923 and not the evidence itself. Generative inpainting is a reasonable tool here,
924 provided its outputs are labelled as generated and are never merged into the

925 evidence record. In *analytical* use, where a reconstruction informs an exam-
926 iner’s interpretation but is not itself tendered, imputed content is a source
927 of interpretive bias whose direction is unknown to the examiner; the risk is
928 real but manageable through disclosure and separate storage of source and
929 reconstruction. Imputed content becomes acute in *evidential* use, where the
930 reconstruction or a conclusion resting on it is put before a finder of fact. A re-
931 constructed license-plate digit that is visually convincing but algorithmically
932 invented shifts the record from what was observed to what was plausibly ob-
933 served, and an opposing expert who identifies the generative step can argue
934 that the disputed characters are artefacts of a training distribution and not
935 of the disk under examination.

936 CSFR targets the third case. Its design cost, lower perceptual quality on
937 severe masks, is only worth paying there. The claim of “no hallucination” is
938 itself bounded: CSFR guarantees that no output pixel is sampled from a prior
939 external to the input fragment, which is a statement about the provenance
940 of imputed values and not a guarantee that they are correct. An imputed
941 pixel derived entirely from observed neighbours can still be wrong, and the
942 HRP results of Section 6.5 show that a method can be maximally consistent
943 with observed statistics while certifying nothing.

944 7.3. What Constraint-Driven Reconstruction Adds

945 A constraint-driven method differs from a black-box generative method in
946 one operationally consequential respect: every reconstructed pixel has a finite,
947 explicit contributing-input set. This lets a laboratory answer audit queries
948 (Figure 2) from a persisted record and produce documentation listing, for
949 every reconstructed pixel, the observed inputs and constraint configuration

950 that determine it. Comparable queries against a generative model require
 951 model-interpretability techniques whose own validity is contested [10, 11].
 952 The classical baselines sit between the two: they are deterministic and their
 953 outputs are in principle reproducible from the input, but their standard im-
 954 plementations persist no such record, which is why they score zero on TS as
 955 released rather than zero in principle (Section 6.8).

956 *7.4. Operational Recommendation*

957 A deployment that follows from the evidence without over-reaching is:
 958 (a) CSFR upstream, with Bounded-Unknown flagging on regions where the
 959 C2–C4 constraints cannot be satisfied and the reconstruction should not be
 960 trusted; (b) a downstream classifier operating on the reconstructed substrate;
 961 (c) human review on every Bounded-Unknown flag and on any reconstruction
 962 tendered evidentially. The downstream benefit is not measured in this paper.
 963 Whether feeding CSFR-reconstructed inputs to a fragment classifier recovers
 964 accuracy or calibration relative to the corrupted inputs is the immediate next
 965 step of this programme (Section 8, item 1). The recommendation rests on the
 966 reconstruction-level properties established here and not on an unmeasured
 967 end-to-end gain.

968 **8. Limitations and Threats to Validity**

969 We state the following limitations explicitly:

- 970 1. **Downstream impact not yet measured.** This paper certifies re-
 971 construction properties; the end-to-end experiment of feeding CSFR-
 972 reconstructed inputs into fragment classifiers and measuring the re-

973 covery of accuracy and calibration has not yet been run and is the
974 immediate next step of this programme.

975 2. **Perceptual realism.** CSFR will not match DL inpainters on PSNR
976 or SSIM at high corruption severities. The framework treats this gap as
977 an intentional design choice reflecting the forensic priority of bounded
978 inference over perceptual fidelity; we make no claim of perceptual parity
979 with diffusion models.

980 3. **Constraint definitions.** The four constraint families C1–C4 are suf-
981 ficient for the corruption modes evaluated in this paper’s sweep. They
982 may not cover all corruption physics encountered in field practice (e.g.,
983 chemical degradation of photographic film). Extension of the constraint
984 family set is left to future work.

985 4. **Solver complexity.** The released first-order solver costs $O(T \cdot N \log N)$
986 per fragment ($T = 600$ Adam iterations, each dominated by the DCT),
987 which is substantially slower than a forward pass through a pre-trained
988 inpainter ($O(N)$ in image area). The reported throughput is there-
989 fore unsuitable for real-time applications; CSFR is intended for offline
990 forensic analysis.

991 5. **Chaotic corruption.** For fragments in which $\Omega = \emptyset$ (no observed
992 pixel) or Ω is so sparse that the RIP assumption on $M\Psi$ is violated,
993 CSFR cannot reconstruct and will emit a Bounded-Unknown flag for
994 the entire fragment.

995 6. **Experimental coverage.** The CSFR-specific result tables in Sec-
996 tions 6.1–6.9 are populated from a real 2-D image-patch $CF \times L$ sweep
997 ($N = 300$ grayscale 64×64 patches per cell carved from DFRWS 2006,

20 cells). Classical PDE inpainting (Telea, Navier–Stokes) [26, 39] and per-image dictionary learning [24, 25] are evaluated under the same harness (Section 6.1); GAN [4, 5] and diffusion [6, 8] baselines remain cited-only because running them would require the learned priors whose evidentiary status the paper contests.

7. **Generalisation beyond JPEG.** The current experimental protocol is anchored on JPEG fragments. Extension to PNG, GIF, raw bitmap, and non-image binary fragment types is straightforward in principle but is not validated in this work.

8. **Dataset-selection bias.** The primary corpus is 300 patches from six DFRWS 2006 images, two of which supply 77% of the patches (Section 5). This concentration makes cluster-level significance testing impossible on that corpus and is the reason the external GovDocs1 corpus, not the primary one, carries the generalisation claim. Both corpora are document- and photograph-heavy JPEG content; performance on other image statistics (medical, satellite, low-light surveillance) is unmeasured.

9. **Implementation and solver bias.** All results reflect one solver realisation, Adam on the penalised surrogate at a fixed 600-iteration budget, and not the idealised constrained programme of Eq. (2). The iterates are reproducible but not certified minimisers. The budget-sensitivity analysis (Section 6.11) shows the CF2 and severe-mask deficits reported in the main tables to be substantially a consequence of this fixed budget being too short for the harder masks, not of the sparsity prior itself: a longer solve recovers most of the gap. The main results are reported

1023 at the audited budget regardless, and this is a bias of the reported
1024 *configuration* that a practitioner can trade against runtime.

1025 10. **Baseline-configuration bias.** The classical baselines run at fixed
1026 settings (OpenCV inpaint radius 3; a 64-atom dictionary with 8-sparse
1027 OMP). A practitioner tuning these per corruption family might nar-
1028 row or widen the gaps reported here. The comparison with generative
1029 methods is structurally asymmetric: they are cited, not run, so no
1030 perceptual number is put head-to-head against CSFR, by design.

1031 11. **Seed count.** Each condition is run at three seeds ($n = 3$), which
1032 controls mask realisation and the dictionary-learning fit. This is ade-
1033 quate for a mean but bounds the precision of the between-seed variance
1034 estimates, which are consequently reported as descriptive rather than
1035 as a basis for inference; the patch dimension ($N = 300$), not the seed
1036 dimension, carries the statistical power in Section 6.2.

1037 12. **From patches to files.** The evaluation reconstructs decoded 64×64
1038 pixel patches. Generalising these results to whole encoded files, to
1039 storage-level fragmentation across non-contiguous clusters, and to the
1040 re-synthesis of a valid encoded bitstream (Section 3) is not established
1041 here and is the principal external-validity threat for operational deploy-
1042 ment.

1043 9. Conclusion

1044 Forensic image reconstruction demands bounded inference and traceable
1045 provenance, where perceptual reconstruction demands only visual realism.
1046 A deep generative reconstruction samples its output pixels from a learned

1047 prior, so it cannot state an error bound or trace a pixel to the recovered
1048 bytes, and that is the core obstacle to its evidential use. Compressed Sensing
1049 Fragment Reconstruction (CSFR) addresses this by posing reconstruction as
1050 a regularised inverse problem under four constraint families. The solver is
1051 deterministic, needs no training data, carries a conditional recovery bound
1052 under the Restricted Isometry Property, and emits a per-pixel provenance
1053 record for chain-of-custody audit.

1054 The evaluation settles the comparison by inference, not point estimates.
1055 CSFR’s quality advantage is unambiguous in one regime, scattered pixel loss,
1056 where a one to one-and-a-half decibel lead holds at every severity, survives
1057 multiplicity correction and source-image clustering, and reproduces on a dis-
1058 joint corpus. Where the corruption is spatially concentrated the ranking
1059 inverts and classical inpainting wins by several decibels, but the budget-
1060 sensitivity analysis shows that deficit to be largely an under-convergence
1061 effect of the deliberately short fixed budget, recoverable, and on burst cor-
1062 ruption reversible, by a longer solve. The practical position is therefore
1063 conditional: under scattered and mild damage an examiner obtains CSFR’s
1064 auditability at no quality cost, and under severe damage the examiner trades
1065 quality for auditability or spends more computation to recover it. What
1066 no evaluated baseline supplies as implemented is the combination of exact
1067 evidence preservation, a stated recovery bound, and a released provenance
1068 record an opposing expert can examine. These properties support auditabil-
1069 ity and reproducibility; admissibility itself remains a determination for a
1070 court.

1071 The broader claim is that evidential reconstruction imposes mathematical

1072 structure that perceptual quality does not, and that this structure is incom-
1073 patible with sampling from a learned prior. Constraint-driven reconstruction
1074 is the regime currently available under which reconstructed image evidence
1075 can meet both the bounded-inference and the traceability requirements of
1076 digital evidence handling. The framework, solver, statistical analysis, and
1077 reproducible protocol are released with the paper.

1078 **Acknowledgements**

1079 The authors thank the FFT-75 dataset maintainers [45] and the main-
1080 tainers of the DFRWS and Nick Mikus challenge corpora. The compressed-
1081 sensing community’s open-source ecosystem, and the PyTorch, NumPy,
1082 SciPy, scikit-image, scikit-learn, and OpenCV projects on which the released
1083 implementation depends, are gratefully acknowledged.

1084 **Ethics Statement**

1085 This study uses only publicly released research corpora (the DFRWS 2006
1086 forensic challenge image and the FFT-75 benchmark); no personal data were
1087 collected and no human participants were involved. Illustrative references to
1088 license-plate or CCTV imagery are hypothetical deployment scenarios, not
1089 experiments performed.

1090 **Declaration of Competing Interest**

1091 The authors declare no competing financial interests or personal relation-
1092 ships that could have appeared to influence the work reported in this paper.

1093 **Funding**

1094 This research received no specific grant from any funding agency in the
1095 public, commercial, or not-for-profit sectors.

1096 **CRedit Author Statement**

1097 **Benjamin Tei Parthey:** Conceptualisation, Methodology, Software, Val-
1098 idation, Formal analysis, Investigation, Data curation, Writing — original
1099 draft, Writing — review and editing. **Yaw Osei Adjei:** Conceptualisation,
1100 Methodology, Supervision, Validation, Writing — review and editing.

1101 **Code and Data Availability**

1102 The complete reproducibility package is released at [https://github.](https://github.com/yoadjei/csfr)
1103 [com/yoadjei/csfr](https://github.com/yoadjei/csfr): the CSFR solver (`src/csfr_reconstruct_2d.py`), the
1104 per-pixel provenance implementation (`src/provenance.py`) with a released
1105 example manifest (`results/provenance_example/`), the sweep and abla-
1106 tion harness (`scripts/run_csfr_sweep.py`), the paired-inference, budget-
1107 sensitivity, benchmark, and frontier-sensitivity scripts (`scripts/run_paired_stats.py`,
1108 `run_budget_sensitivity.py`, `run_bench.py`, `run_frontier_sensitivity.py`),
1109 the patch-extraction protocol (`scripts/extract_patches.py`), the ex-
1110 periment configurations (`configs/`), the patch corpora with per-patch
1111 source-image cluster labels and extraction manifests (`data/`), all per-cell,
1112 per-seed metric artefacts and the analysis outputs (`results/`), figure-
1113 generation scripts, and unit/smoke tests. Pinned dependency versions are in
1114 `requirements.txt`; each result directory’s `run_metadata.json` records the

environment that produced it. The repository will be archived to Zenodo
with a version DOI on acceptance.

References

- [1] B. Carrier, File System Forensic Analysis, Addison-Wesley Professional, 2005.
- [2] E. Casey, Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet, 3rd Edition, Academic Press, 2011.
- [3] S. L. Garfinkel, Carving contiguous and fragmented files with fast object validation, Digital Investigation 4 (2007) S2–S12. doi:10.1016/j.diin.2007.06.017.
- [4] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, T. S. Huang, Generative image inpainting with contextual attention, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 5505–5514. doi:10.1109/CVPR.2018.00577.
- [5] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, V. Lempitsky, Resolution-robust large mask inpainting with Fourier convolutions, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022, pp. 2149–2159. doi:10.1109/WACV51458.2022.00323.
- [6] J. Ho, A. Jain, P. Abbeel, Denoising diffusion probabilistic models, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 33,

1137 2020, pp. 6840–6851.
1138 URL <https://arxiv.org/abs/2006.11239>

1139 [7] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-
1140 resolution image synthesis with latent diffusion models, in: Proceedings
1141 of the IEEE/CVF Conference on Computer Vision and Pattern Recogni-
1142 tion (CVPR), 2022, pp. 10684–10695. doi:10.1109/CVPR52688.2022.
1143 01042.

1144 [8] A. Lugmayr, M. Danelljan, A. Romero, F. Yu, R. Timofte, L. Van Gool,
1145 RePaint: Inpainting using denoising diffusion probabilistic models, in:
1146 Proceedings of the IEEE/CVF Conference on Computer Vision and
1147 Pattern Recognition (CVPR), 2022, pp. 11461–11471. doi:10.1109/
1148 CVPR52688.2022.01117.

1149 [9] C. Zheng, T.-J. Cham, J. Cai, Pluralistic image completion, in: Proceed-
1150 ings of the IEEE/CVF Conference on Computer Vision and Pattern
1151 Recognition (CVPR), 2019, pp. 1438–1447. doi:10.1109/CVPR.2019.
1152 00153.

1153 [10] S. Bhadra, V. A. Kelkar, F. J. Brooks, M. A. Anastasio, On hallu-
1154 cinations in tomographic image reconstruction, IEEE Transactions on
1155 Medical Imaging 40 (11) (2021) 3249–3260. doi:10.1109/TMI.2021.
1156 3077857.

1157 [11] G. Horsman, Sources of error in digital forensics, Forensic Science In-
1158 ternational: Digital Investigation 48 (2024) 301693. doi:10.1016/j.
1159 fsidi.2024.301693.

- 1160 [12] Ç. Arslan, S. Ata, The reliability of digital evidence in criminal proceed-
1161 ings and the potential utilization of artificial intelligence in the evidence
1162 evaluation process, in: 12th International Symposium on Digital Foren-
1163 sics and Security (ISDFS), 2024, pp. 1–4. doi:10.1109/ISDFS60797.
1164 2024.10527346.
- 1165 [13] United States Supreme Court, Daubert v. Merrell Dow Pharmaceuticals,
1166 Inc., 509 U.S. 579, leading case establishing admissibility standard for
1167 scientific expert testimony in US federal courts (1993).
- 1168 [14] R v Bowden [1999] 4 All ER 43, Court of Appeal (Criminal Division),
1169 leading UK case on digital forensics and evidence admissibility (1999).
- 1170 [15] A. Pal, N. Memon, The evolution of file carving, IEEE Signal Processing
1171 Magazine 26 (2) (2009) 59–71. doi:10.1109/MSP.2008.937841.
- 1172 [16] E. J. Candès, J. K. Romberg, T. Tao, Robust uncertainty princi-
1173 ples: Exact signal recovery from highly incomplete frequency informa-
1174 tion, IEEE Transactions on Information Theory 52 (2) (2006) 489–509.
1175 doi:10.1109/TIT.2005.862083.
- 1176 [17] D. L. Donoho, Compressed sensing, IEEE Transactions on Information
1177 Theory 52 (4) (2006) 1289–1306. doi:10.1109/TIT.2006.871582.
- 1178 [18] E. J. Candès, J. K. Romberg, T. Tao, Stable signal recovery from incom-
1179 plete and inaccurate measurements, Communications on Pure and Ap-
1180 plied Mathematics 59 (8) (2006) 1207–1223. doi:10.1002/cpa.20124.
- 1181 [19] E. J. Candès, The restricted isometry property and its implications for

- 1182 compressed sensing, *Comptes Rendus Mathematique* 346 (9–10) (2008)
1183 589–592. doi:10.1016/j.crma.2008.03.014.
- 1184 [20] S. Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way*, 3rd
1185 Edition, Academic Press, 2009.
- 1186 [21] L. I. Rudin, S. Osher, E. Fatemi, Nonlinear total variation based noise
1187 removal algorithms, *Physica D: Nonlinear Phenomena* 60 (1–4) (1992)
1188 259–268. doi:10.1016/0167-2789(92)90242-F.
- 1189 [22] A. N. Tikhonov, V. Y. Arsenin, *Solutions of Ill-Posed Problems*, Winston
1190 & Sons, 1977.
- 1191 [23] M. Elad, M. Aharon, Image denoising via sparse and redundant repre-
1192 sentations over learned dictionaries, *IEEE Transactions on Image Pro-*
1193 *cessing* 15 (12) (2006) 3736–3745. doi:10.1109/TIP.2006.881969.
- 1194 [24] J. Mairal, F. Bach, J. Ponce, G. Sapiro, Online dictionary learning for
1195 sparse coding, *Proceedings of the International Conference on Machine*
1196 *Learning (ICML)* (2009) 689–696doi:10.1145/1553374.1553463.
- 1197 [25] M. Aharon, M. Elad, A. Bruckstein, K-SVD: An algorithm for designing
1198 overcomplete dictionaries for sparse representation, *IEEE Transactions*
1199 *on Signal Processing* 54 (11) (2006) 4311–4322. doi:10.1109/TSP.2006.
1200 881199.
- 1201 [26] M. Bertalmio, G. Sapiro, V. Caselles, C. Ballester, Image inpainting,
1202 *Proceedings of ACM SIGGRAPH* (2000) 417–424doi:10.1145/344779.
1203 344972.

- 1204 [27] A. Criminisi, P. Pérez, K. Toyama, Region filling and object removal by
1205 exemplar-based image inpainting, *IEEE Transactions on Image Process-*
1206 *ing* 13 (9) (2004) 1200–1212. doi:10.1109/TIP.2004.833105.
- 1207 [28] C. Corneanu, R. Gadde, A. M. Martinez, Latentpaint: Image in-
1208 painting in latent space with diffusion models, in: *IEEE/CVF Win-*
1209 *ter Conference on Applications of Computer Vision (WACV)*, 2024.
1210 doi:10.1109/WACV57701.2024.00428.
- 1211 [29] A. Grechka, G. Couairon, M. Cord, Gradpaint: Gradient-guided inpaint-
1212 ing with diffusion models, *Computer Vision and Image Understanding*
1213 240 (2024) 103928. doi:10.1016/j.cviu.2024.103928.
- 1214 [30] R. Poisel, S. Tjoa, Forensics investigations of multimedia data: A review
1215 of the state-of-the-art, in: *Sixth International Conference on IT Security*
1216 *Incident Management and IT Forensics (IMF)*, 2011, pp. 48–61. doi:
1217 10.1109/IMF.2011.14.
- 1218 [31] N. Mikus, An analysis of disc carving techniques, in: *Naval Postgraduate*
1219 *School Technical Report*, 2005.
1220 URL <http://dfitt.sourceforge.net/>
- 1221 [32] W. C. Calhoun, D. Coles, Predicting the types of file fragments, *Digital*
1222 *Investigation* 5 (2008) S14–S20. doi:10.1016/j.diin.2008.05.005.
- 1223 [33] C. E. Shannon, A mathematical theory of communication, *The Bell*
1224 *System Technical Journal* 27 (3) (1948) 379–423. doi:10.1002/j.
1225 1538-7305.1948.tb00917.x.

- 1226 [34] Scientific Working Group on Digital Evidence (SWGDE), Best practices
1227 for digital and multimedia evidence, SWGDE DocumentsVersion 4.0
1228 (2023).
1229 URL [https://www.swgde.org/documents/
1230 published-complete-listing/](https://www.swgde.org/documents/published-complete-listing/)
- 1231 [35] H. Farid, Image forgery detection, IEEE Signal Processing Magazine
1232 26 (2) (2009) 16–25. doi:10.1109/MSP.2008.931079.
- 1233 [36] M. Tivnan, S. Yoon, Z. Chen, X. Li, D. Wu, Q. Li, Hallucination
1234 index: An image quality metric for generative reconstruction mod-
1235 els, in: Medical Image Computing and Computer Assisted Intervention (MICCAI), Lecture Notes in Computer Science, 2024, pp. 449–458.
1236 doi:10.1007/978-3-031-72117-5_42.
1237
- 1238 [37] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in:
1239 3rd International Conference on Learning Representations (ICLR), 2015.
1240 URL <https://arxiv.org/abs/1412.6980>
- 1241 [38] S. Garfinkel, P. Farrell, V. Roussev, G. Dinolt, Bringing science to digi-
1242 tal forensics with standardized forensic corpora, Digital Investigation 6
1243 (2009) S2–S11. doi:10.1016/j.diin.2009.06.016.
- 1244 [39] A. Telea, An image inpainting technique based on the fast marching
1245 method, Journal of Graphics Tools 9 (1) (2004) 23–34. doi:10.1080/
1246 10867651.2004.10487596.
- 1247 [40] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, Image quality
1248 assessment: From error visibility to structural similarity, IEEE Trans-

- actions on Image Processing 13 (4) (2004) 600–612. doi:10.1109/TIP.
2003.819861.
- [41] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics
Bulletin 1 (6) (1945) 80–83. doi:10.2307/3001968.
- [42] S. Holm, A simple sequentially rejective multiple test procedure, Scan-
dinavian Journal of Statistics 6 (2) (1979) 65–70.
URL <https://www.jstor.org/stable/4615733>
- [43] J. Cohen, Statistical Power Analysis for the Behavioral Sciences, 2nd
Edition, Lawrence Erlbaum Associates, 1988.
- [44] N. Cliff, Dominance statistics: Ordinal analyses to answer ordinal ques-
tions, Psychological Bulletin 114 (3) (1993) 494–509. doi:10.1037/
0033-2909.114.3.494.
- [45] G. Mittal, S. D. Saha, C. Castillo, FFT-75: File fragment type dataset
— 75 classes, IEEE DataPort (2020).
URL [https://ieee-dataport.org/open-access/
file-fragment-type-fft-75-dataset](https://ieee-dataport.org/open-access/file-fragment-type-fft-75-dataset)